

On Einstein's Theory of Gravitation

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Preface

This work is inspired by and based on the lectures delivered by Dr. Kinjalk Lochan in the spring semester (January to May) of 2025 on general relativity and cosmology at IISER Mohali. Here, the work has been lucidly presented, covering most of the subtleties, but still, it may be possible that certain errors crept in and remain unnoticed. It is aimed at someone desirous of getting an overview of the topic. I am thankful to him for teaching the topic. I am also thankful for the @PhysicsUnsimplified YouTube creator for his lecture videos on covariant and Lie derivatives. I am also thankful to my colleague and friend, Mr. Angring Engti, for providing lecture notes, particularly on the topics of gravitational waves and others. I am thankful to all my friends and family members for their constant support.

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Chapter 1

Introduction

Okay! Now, we introduce this topic. Look, I hope that from the special theory of relativity lectures, you would have realised the fact that our daily life (as of 2025) does not significantly perceive the effect of special relativity. Regularly, one cites the expression, $\frac{v}{c} \ll 1$ when describing physics in ordinary life. But why? Isn't that the ratio of two speeds? Or why this particular ratio? Why can't we have a ratio of two lengths? The ratio of two masses or the ratio of two energies, etc. Such that they all by themselves distinguish ordinary life from relativistic life. People may say that the description of Special relativity generally deals with incorporating an inertial frame moving relative to another inertial frame with a velocity, say v , which is why taking the ratio between v and absolute universal velocity c (maximum speed of propagation of the signal in nature, aka speed of light) seems logical. You see, wouldn't that have sounded weird if the frame slid with mass m or energy E ?

Fortunately, people have an incomplete understanding of this. You see, we can also have a ratio of energies as $\frac{2\varepsilon}{E_0} \ll 1$, where $E_0 = mc^2$ and ε is ordinary energy in correspondence to v . Similar things can be said about momentum. The only thing one should have in mind is that there has to be a universal/absolute "entity" that has to be present in the 'ratio' expression. Now, you know, when you will be dealing with quantum mechanics or high energy physics, what sort of limiting condition will be advantageous and while indulging in relativistic mechanics calculations, then why should you not always keep in mind about v/c ratio only? You can have any desired form of such a dimensionless ratio, which helps you distinguish non-relativistic (ordinary) from relativistic regimes. Since we are interested in general relativity, let me put that again, "it's the general theory of relativity", and mostly hangs with large-scale pieces of stuff, unlike quantum-scale or high-energy, therefore, could you guess taking whose ratio (physical quantity) that may be turned out to be beneficial? I think "length" is a good choice. It's a viva question about how length and velocity are connected in the description of gravity. Finally, we see $\frac{R_s}{r}$ is a good fit unless you are keeping track of underlying analogies. Anyway, where R_s is the Schwarzschild radius and r is the usual ordinary radius (length). Note carefully that R_s is a function of M , the mass of the source body, just like how E_0 depends on mass m . Therefore, $\frac{R_s}{r} \ll 1$ is a non-relativistic limiting condition.

Your professor should have told you about the Eötvös number and the following expression

$$\frac{m_g}{m_i} - 1 \approx 0$$

with Average sensitivity of $\pm 10^{-16}$.

If you rewrite all these discussed ratios while respecting that it is impossible to get $v = c$. Then, we seemingly have these forms.

$$\begin{aligned}\frac{v}{c} - 1 &\ll 0 \\ \frac{2\varepsilon}{E_0} - 1 &\ll 0 \\ \frac{R_s}{r} - 1 &\ll 0\end{aligned}$$

My question: Why do they look similar to the Eötvös number but in a complementary sense? By complementary, I mean that we are more sure $m_g = m_i$ always, but $v \neq c$ always in any case. In short, nature tends to make us believe that inertial mass is equivalent to gravitation mass, but does not allow us to accelerate the velocity of an object to reach the speed of light.

1.1 Postulates

Let's partially review the theory of special relativity. At the end of the day, you have realised that basically, A. Einstein wanted to keep the principle of relativity intact, even though the Galilean transformation, which is grounded upon Newton's laws of motion, bore severe foundational tremors. His remarkable idea is that the speed of light must remain invariant or constant irrespective of observers in any inertial frame, which shatters the Galilean transformation completely and subsequently, Newton's laws, at the cost of correcting the mathematical form of mass, momentum, forces, Etc. Along with motivating a major shift in the understanding of space-time, Newton's second law still had an impact smoothly. And, involving a novel set of transformations which respect both the above postulates of special relativity. One should never forget that Maxwell's laws, which have always been compatible with relativity, have dramatically passed their toughest trial. A deep glance over these would make us realise that A. Einstein never wanted to prove, disprove or make a change in the foundation of classical mechanics or mathematical form, but in fact, he wanted to keep the foundational form intact with the necessary corrections.

In general relativity, we again find that a similar procedure for theoretical investigations was unwrapped. His second postulate (the first one being the constancy of the speed of light) about the physics of nature, or the first postulate of general relativity, is the principle of equivalence. Thus, one can say that the principle of relativity, the constancy of the speed of light, and the principle of equivalence together form the basis of the theory of general relativity. The Eötvös number should be zero, which is known as the weak principle of equivalence. Note that a weak principle is weak in assumption-wise only, but way stronger in its implication than its strongly named version.

In the end, in the case of free particles, A. Einstein aims to keep the following expression intact.

$$\frac{d^2x}{dt^2} = 0$$

Chapter 2

Newton's Second Law LHS

2.1 The Geodesic Equation

Let $\zeta^\mu(x^\alpha)$ be a “special” spacetime four coordinates. And, as indicated, it is a function of the general four-coordinates x^α . We consider τ a proper time: time measured in the observer self-frame (an event with no change in spatial position $\delta x = 0$; a Lorentz-invariant quantity with the value being the smallest of all).

We claim that gravity disappears in that special frame (Local inertial: LIF), which is described through ζ^μ . Therefore, with the left-hand side (LHS) of Newton's 2nd law in relativistic settings as written below, we work for the objective: can one retain explicit form after a general coordinate transformation or not?

$$\frac{d^2}{d\tau^2}\zeta^\mu(x^\alpha) = \frac{d}{d\tau} \left(\frac{d}{d\tau}\zeta^\mu(x^\alpha) \right) = 0$$

using the chain rule,

$$d\zeta^\mu = \frac{\partial \zeta^\mu}{\partial x^\alpha} dx^\alpha$$

[Don't forget that the above-connected term (in red) is just a general example of Jacobian or, in the case of special relativity, it is the Lorentz transformation (LT) marked by Λ_α^μ].

Or,

$$\frac{d\zeta^\mu}{d\tau} = \frac{\partial \zeta^\mu}{\partial x^\alpha} \frac{dx^\alpha}{d\tau}$$

Or,

$$\frac{d}{d\tau} \left(\frac{d}{d\tau}\zeta^\mu(x^\alpha) \right) = \frac{d}{d\tau} \left(\frac{\partial \zeta^\mu}{\partial x^\alpha} \frac{dx^\alpha}{d\tau} \right) = 0$$

Using Leibniz's product rule,

$$\implies \frac{d}{d\tau} \left(\frac{\partial \zeta^\mu}{\partial x^\alpha} \right) \frac{dx^\alpha}{d\tau} + \frac{d}{d\tau} \left(\frac{dx^\alpha}{d\tau} \right) \frac{\partial \zeta^\mu}{\partial x^\alpha} = 0$$

Upon a few rearrangements (which include the action of Kronecker delta δ_μ^β , multiplying both sides by the inverse matrix $M^{-1} \equiv M_\mu^\beta = \frac{\partial x^\beta}{\partial \zeta^\mu}$ followed by a transposition,)

$$\frac{d^2 x^\beta}{d\tau^2} + M_\mu^\beta \frac{dM_\alpha^\mu}{d\tau} \frac{dx^\alpha}{d\tau} = 0$$

Note: Don't forget the root, at every step of our manipulation, that has always been the LHS of Newton's 2nd law, still some physicists *will* love to call the upcoming resulting version of the given equation a Geodesics equation. Just weird folks! Anyway, at this stage of manipulation, we should note a few Insights and a few Questions, listed as follows:

- Q: We already know how μ -coordinates and α -coordinates are interconnected, but How $c\tau$ -coordinate (temporal part of very specialized frame) and α are interconnected?
- I: Note that if either M_α^μ or x^α (rest) happens to be time-independent, then we would have shown $\frac{d^2}{d\tau^2}$ secures its form under the General Coordinate Transformation ($\zeta^\mu \rightarrow x^\alpha$). Then we have achieved the objective. Alas! This does not generally hold.

Thus, answering the above question, we introduce x^λ four coordinates, such that $x^{\lambda=0}$ maps with $c\tau$ for the light-like case. As a result, we see the following changes (again due to the chain rule) with the above under-constructed Geodesics equation,

$$\frac{d^2 x^\beta}{d\tau^2} + \left[M_\mu^\beta \frac{dM_\alpha^\mu}{dx^\lambda} \frac{dx^\lambda}{d\tau} \frac{dx^\alpha}{d\tau} \right] = 0$$

If you are still thinking about why we introduced λ four coordinates, then I suggest that you consider $M_\alpha^\mu = M_\alpha^\mu(x^\lambda)$, instead of just looking at specialised $M_\alpha^\mu(\tau)$. Basically, we are upgrading the domain of M_α^μ . Let me recall the objective: we are doing a general coordinate transformation and at this stage, we can see that “technically” gravity is indeed absent, but additionally, from the above, we see that we have carried an extra geometrical factor [...], spacetime being its ingredient in the course of the required transformation. At this early stage, can gravity, being a spacetime construct, be a valid guess or not? We will realise the truth in later sections.

You may agree with me if I claim that we have satisfactorily interconnected everything to its fullest up to this stage. Mathematicians also love compactification, they defined Christoffel's Connection:

$$\Gamma_{\lambda\alpha}^\beta = M_\mu^\beta \frac{dM_\alpha^\mu}{dx^\lambda} = \left(\frac{\partial x^\beta}{\partial \zeta^\mu} \right) \frac{d}{dx^\lambda} \left(\frac{d\zeta^\mu}{dx^\alpha} \right)$$

If I convey to you that the metric of flat spacetime (Minkowski metric) is $\eta_{\mu\nu}$ in Minkowski geometry. And, when one deals with the geometry of curved spacetime (Riemannian geometry), it obviously requires a useful upgrade to be capable of withstanding “curviness”. We call that $g_{\alpha\beta}$. It is common sense to notice that a curve locally seems flat.

Therefore, $g_{\alpha\beta} = \eta_{\mu\nu} K_{,\alpha,\beta}^{\mu\nu}$ should be true. Anyway, we call $g_{\alpha\beta}$ a Metric Tensor, and it is a geometrical object. As the flat differential spacetime interval (Lorentz invariant) in Minkowski geometry is $ds^2 = \eta_{\mu\nu} d\zeta^\mu d\zeta^\nu$, similarly, the curved differential spacetime interval in Riemannian geometry should be $ds^2 = g_{\alpha\beta} dx^\alpha dx^\beta$. It is an exercise to find out the connecting factor $K_{,\alpha,\beta}^{\mu\nu}$, and then you shall be motivated to relate Christoffel's Connection with the Metric tensor, once you hopefully dissert about $\frac{d\zeta^\mu}{dx^\alpha}$ appearance. The answer should be Christoffel's connection (explicitly Christoffel symbol of the second kind) as a function of the metric tensor, as,

$$\Gamma_{\lambda\alpha}^\beta = \frac{1}{2} g^{\beta\rho} (-g_{\lambda\alpha,\rho} + g_{\rho\lambda,\alpha} + g_{\rho\alpha,\lambda})$$

Now you know two expressions of Christoffel's connection.

Finally, using the above understanding, we have fully constructed the Geodesics equation as,

$$\frac{d^2 x^\beta}{d\tau^2} + \Gamma_{\lambda\alpha}^\beta \frac{dx^\lambda}{d\tau} \frac{dx^\alpha}{d\tau} = 0$$

Also, recall that this is just a very manipulated (generalised) expression of the LHS of Newton's 2nd law. Still, you can argue with weird folks. Notice that one may also regard it as an expression describing a free particle (no interaction, RHS is zero), but more explicitly, its solution should depict the trajectory of a free particle under consideration in curved spacetime. If you had multiplied both sides by inertial mass, you would have got an enhanced version of $m\ddot{x} = 0$, more correctly, $m\ddot{x} = -m\mathcal{O}$. If Newton had seen that, he would have remarked about $\mathcal{O}$ being related to the respective "innate force". We seem to have failed in our intention of getting $\ddot{x}^\alpha = 0$ if given $\ddot{\zeta}^\mu = 0$. Nevertheless, we found that the geodesic equation is a much better form that retains its form under general coordinate transformation. Initially, if we had based our objective on keeping the Geodesic equation intact, our mission would have succeeded. Anyway, failure teaches more than victory.

2.1.1 Small Exercise

Question 1: Derive the form of $K_{\alpha,\beta}^{\mu\nu}$, that relates $g_{\alpha\beta}$ with $\eta_{\mu\nu}$.

It may be possible that you solved the exercise a way before, as when being instructed, and for those who did not do so, then, here is the solution:

Given:

$$\begin{aligned} ds^2 &= \eta_{\mu\nu} d\zeta^\mu d\zeta^\nu \\ &= \eta_{\mu\nu} \left(\frac{\partial \zeta^\mu}{\partial x^\alpha} dx^\alpha \right) \left(\frac{\partial \zeta^\nu}{\partial x^\beta} dx^\beta \right) \\ &= \eta_{\mu\nu} \left(\frac{\partial \zeta^\mu}{\partial x^\alpha} \frac{\partial \zeta^\nu}{\partial x^\beta} \right) dx^\alpha dx^\beta \end{aligned}$$

Let's identify $K_{\alpha,\beta}^{\mu\nu} = \frac{\partial \zeta^\mu}{\partial x^\alpha} \frac{\partial \zeta^\nu}{\partial x^\beta}$, that makes $g_{\alpha\beta} = \eta_{\mu\nu} K_{\alpha,\beta}^{\mu\nu}$ then,

$$ds^2 = g_{\alpha\beta} dx^\alpha dx^\beta$$

This becomes the line element of the curved spacetime manifold.

Question 2: If the trajectory of a free particle in Cartesian coordinates $\zeta^m(t, x, y)$ is a straight line, we are asked to find its trajectory in polar coordinates $x^a(t', r, \theta)$. note: $c = 1$. If ζ^m are related to τ by multiplicative constants c_0, c_1 and c_2 respectively i.e., $x^l(c_0\tau, c_1\tau, c_2\tau)$.

Answer: We know $\frac{d^2 \zeta^m}{d\tau^2} = 0$. Now, to find the trajectory of the free particle, we should solve the following geodesic equation,

$$\frac{d^2 x^\beta}{d\tau^2} + \Gamma_{la}^\beta \frac{dx^l}{d\tau} \frac{dx^a}{d\tau} = 0$$

Let's find Γ_{la}^β , we use the Matrix form, $\left(\Gamma_{la}^\beta = M_m^\beta \frac{dM_a^m}{dx^l} \right)$ of it, as finding both Jacobian $J_{x^a}(\zeta^m)$ or simply M_a^m and inverse M_m^β . We use the relation $t' = t$, $r = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1} \left(\frac{y}{x} \right)$

$$M_a^m = \begin{bmatrix} \frac{\partial t'}{\partial t} & \frac{\partial t'}{\partial x} & \frac{\partial t'}{\partial y} \\ \frac{\partial r}{\partial t} & \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial t} & \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{x}{\sqrt{x^2+y^2}} & \frac{y}{\sqrt{x^2+y^2}} \\ 0 & \frac{-y}{x^2+y^2} & \frac{x}{x^2+y^2} \end{bmatrix}$$

Finding $\frac{dM_a^m}{dx^l}$, it seems the result would be 3D matrix, third extension due to x^l . Thus, we find slice by slice as,

$$\frac{dM_a^m}{dx^0} = \begin{bmatrix} \frac{d}{dx^0}(1) & 0 & 0 \\ 0 & \frac{d}{dx^0} \left(\frac{x}{\sqrt{x^2+y^2}} \right) & \frac{d}{dx^0} \left(\frac{y}{\sqrt{x^2+y^2}} \right) \\ 0 & \frac{d}{dx^0} \left(\frac{-y}{x^2+y^2} \right) & \frac{d}{dx^0} \left(\frac{x}{x^2+y^2} \right) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \frac{d}{dx^0} \left(\frac{-y}{x^2+y^2} \right) & \frac{d}{dx^0} \left(\frac{x}{x^2+y^2} \right) \end{bmatrix}$$

A more quick method is to use $t = c_0\tau \equiv x^0$, $x = c_1\tau \equiv x^1$ and $y = c_2\tau \equiv x^2$ gives,

$$\frac{dM_a^m}{dx^0} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \frac{1}{c_0} \frac{d}{d\tau} \left(\frac{-c_2/\tau}{c_1^2+c_2^2} \right) & \frac{1}{c_0} \frac{d}{d\tau} \left(\frac{c_1/\tau}{c_1^2+c_2^2} \right) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \frac{c_2}{c_0(c_1^2+c_2^2)} \left(\frac{1}{\tau^2} \right) & -\frac{c_1}{c_0(c_1^2+c_2^2)} \left(\frac{1}{\tau^2} \right) \end{bmatrix}$$

similarly,

$$\frac{dM_a^m}{dx^1} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \frac{c_2}{c_1(c_1^2+c_2^2)} \left(\frac{1}{\tau^2} \right) & -\frac{1}{(c_1^2+c_2^2)} \left(\frac{1}{\tau^2} \right) \end{bmatrix}$$

and

$$\frac{dM_a^m}{dx^2} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \frac{1}{(c_1^2+c_2^2)} \left(\frac{1}{\tau^2} \right) & -\frac{c_1}{c_2(c_1^2+c_2^2)} \left(\frac{1}{\tau^2} \right) \end{bmatrix}$$

Also, the inverse matrix is,

$$J_{\zeta^m}(x^\beta) = M_m^\beta = \begin{bmatrix} \frac{\partial t}{\partial x^{\beta_0}} & \frac{\partial t}{\partial x^{\beta_1}} & \frac{\partial t}{\partial x^{\beta_2}} \\ \frac{\partial x}{\partial x^{\beta_0}} & \frac{\partial x}{\partial x^{\beta_1}} & \frac{\partial x}{\partial x^{\beta_2}} \\ \frac{\partial y}{\partial x^{\beta_0}} & \frac{\partial y}{\partial x^{\beta_1}} & \frac{\partial y}{\partial x^{\beta_2}} \end{bmatrix}$$

We also need to find, $\frac{dx^l}{d\tau} \frac{dx^a}{d\tau}$ as,

$$\frac{dx^l}{d\tau} \frac{dx^a}{d\tau} = \left(\frac{\partial c_0\tau}{\partial \tau} + \frac{\partial c_1\tau}{\partial \tau} + \frac{\partial c_2\tau}{\partial \tau} \right) \frac{dx^a}{d\tau} = (c_0 + c_1 + c_2) \left(\frac{\partial t'}{\partial \tau} + \frac{\partial r}{\partial \tau} + \frac{\partial \theta}{\partial \tau} \right)$$

Using, $t' = t = c_0\tau$, $r = \sqrt{x^2 + y^2} = \tau\sqrt{c_1^2 + c_2^2}$ and $\theta = \tan^{-1} \left(\frac{y}{x} \right) = \tan^{-1} \left(\frac{c_2}{c_1} \right)$ then,

$$\Rightarrow \frac{dx^l}{d\tau} \frac{dx^a}{d\tau} = (c_0 + c_1 + c_2) \left(c_0 + \sqrt{c_1^2 + c_2^2} + 0 \right)$$

Now, we have found the required parts of the Geodesic equation. Upon combining them (note: this is not the correct way of putting things back together, we are keeping them for the sake of doing it), we have,

$$\frac{d^2 x^\beta}{d\tau^2} + \begin{bmatrix} \frac{\partial t}{\partial x^{\beta_0}} & \frac{\partial t}{\partial x^{\beta_1}} & \frac{\partial t}{\partial x^{\beta_2}} \\ \frac{\partial x}{\partial x^{\beta_0}} & \frac{\partial x}{\partial x^{\beta_1}} & \frac{\partial x}{\partial x^{\beta_2}} \\ \frac{\partial y}{\partial x^{\beta_0}} & \frac{\partial y}{\partial x^{\beta_1}} & \frac{\partial y}{\partial x^{\beta_2}} \end{bmatrix} \left(\frac{dM_a^m}{dx^0}; \frac{dM_a^m}{dx^1}; \frac{dM_a^m}{dx^2} \right) \begin{bmatrix} c_0^2 & c_0\sqrt{c_1^2+c_2^2} & 0 \\ c_1c_0 & c_1\sqrt{c_1^2+c_2^2} & 0 \\ c_2c_0 & c_2\sqrt{c_1^2+c_2^2} & 0 \end{bmatrix} = 0$$

Now, for the values of β , we will get 3 differential equations.

$$\frac{d^2 x^0}{d\tau^2} = -c_0^2\Gamma_{00}^0 - c_0\sqrt{c_1^2 + c_2^2}\Gamma_{01}^0 - 0\Gamma_{02}^0 - c_1c_0\Gamma_{10}^0 - c_1\sqrt{c_1^2 + c_2^2}\Gamma_{11}^0 - 0\Gamma_{12}^0 - c_2c_0\Gamma_{20}^0 - c_2\sqrt{c_1^2 + c_2^2}\Gamma_{21}^0 - 0\Gamma_{22}^0$$

$$\frac{d^2 x^1}{d\tau^2} = -c_0^2\Gamma_{00}^1 - c_0\sqrt{c_1^2 + c_2^2}\Gamma_{01}^1 - 0\Gamma_{02}^1 - c_1c_0\Gamma_{10}^1 - c_1\sqrt{c_1^2 + c_2^2}\Gamma_{11}^1 - 0\Gamma_{12}^1 - c_2c_0\Gamma_{20}^1 - c_2\sqrt{c_1^2 + c_2^2}\Gamma_{21}^1 - 0\Gamma_{22}^1$$

$$\frac{d^2 x^2}{d\tau^2} = -c_0^2 \Gamma_{00}^2 - c_0 \sqrt{c_1^2 + c_2^2} \Gamma_{01}^2 - 0 \Gamma_{02}^2 - c_1 c_0 \Gamma_{10}^2 - c_1 \sqrt{c_1^2 + c_2^2} \Gamma_{11}^2 - 0 \Gamma_{12}^2 - c_2 c_0 \Gamma_{20}^2 - c_2 \sqrt{c_1^2 + c_2^2} \Gamma_{21}^2 - 0 \Gamma_{22}^2$$

Upon simplification, we have,

$$\frac{d^2 x^0}{d\tau^2} = -c_0 (c_0 \Gamma_{00}^0 + c_1 \Gamma_{10}^0 + c_2 \Gamma_{20}^0) - \sqrt{c_1^2 + c_2^2} (c_0 \Gamma_{01}^0 + c_1 \Gamma_{11}^0 + c_2 \Gamma_{21}^0)$$

$$\frac{d^2 x^1}{d\tau^2} = -c_0 (c_0 \Gamma_{00}^1 + c_1 \Gamma_{10}^1 + c_2 \Gamma_{20}^1) - \sqrt{c_1^2 + c_2^2} (c_0 \Gamma_{01}^1 + c_1 \Gamma_{11}^1 + c_2 \Gamma_{21}^1)$$

$$\frac{d^2 x^2}{d\tau^2} = -c_0 (c_0 \Gamma_{00}^2 + c_1 \Gamma_{10}^2 + c_2 \Gamma_{20}^2) - \sqrt{c_1^2 + c_2^2} (c_0 \Gamma_{01}^2 + c_1 \Gamma_{11}^2 + c_2 \Gamma_{21}^2)$$

To resolve them further, we need to evaluate $\Gamma_{l0}^\beta = M_m^\beta \frac{dM_0^m}{dx^l}$ and $\Gamma_{l1}^\beta = M_m^\beta \frac{dM_1^m}{dx^l}$ only. Let's put this idea in mind that M_{row}^{column} , then using the above calculated matrices, we see,

$$\Gamma_{l0}^\beta = M_0^\beta \frac{dM_0^0}{dx^l} + M_1^\beta \frac{dM_0^1}{dx^l} + M_2^\beta \frac{dM_0^2}{dx^l} = M_0^\beta \{0; 0; 0\} + M_1^\beta \{0; 0; 0\} + M_2^\beta \{0; 0; 0\} = 0$$

similarly,

$$\Gamma_{l1}^\beta = M_0^\beta \frac{dM_1^0}{dx^l} + M_1^\beta \frac{dM_1^1}{dx^l} + M_2^\beta \frac{dM_1^2}{dx^l} = M_0^\beta \{0; 0; 0\} + M_1^\beta \{0; 0; 0\} + M_2^\beta \{0; 0; 0\} = 0$$

Thus, we finally see,

$$\frac{d^2 x^\beta}{d\tau^2} = 0$$

Hence, in $x^a(t', r, \theta)$ spacetime, the trajectory of the free particle will be a straight line since the slope $\frac{dx^\beta}{d\tau} = k$ is constant or the solution is $x^\beta = k\tau + \tau_0$.

One does not always need to go through the above-discussed method. Rather, if one tries to involve the concept of **metric** to find the corresponding non-zero Christoffel symbol, the task can be accomplished easily.

Chapter 3

Metric, Metric, open your mouth

3.1 Metric and transformation

With a certain level of awakening, one can show that the space we live in is curved and time is weird; one is also intermixed, and altogether the spacetime is curved. Gravity- misspelt as a force in fact exists as a result of curvature due to the presence of mass and energy in spacetime.

You know that a calibrated measuring ruler cannot measure the length of a bowl's concave arc, but can determine the diametrical distance of a given hemispherical bowl. The ruler, being a metric instrument, works fine on flat surfaces or Euclidean geometry. And since we need to study curved "things," so need to recall Riemannian geometry, which also motivates us to change the metric instrument to a suitable one. From special relativity, we found that to study the hyperbolic geometry of spacetime, we need the Minkowski metric. Here, in the case of general relativity, we find that gravity is a curvature in this spacetime manifold due to the presence of energy or mass. Due to that, the spacetime manifold would be experiencing lots of disturbance, strain, vortex, stress or shearing, etc, in the corresponding way how matter or energy plays. Therefore, neither the Euclidean metric (identity) nor the Minkowski metric (hyperbolic signatures) would work. A slower and detailed intuitive explanation is given below.

3.1.1 Ruler or Metric

We know there can be a different class of surfaces. Some may be an absolute plane, others may have a mixture of valleys and peaks like a landscape. Some may be hyperbolic surfaces, etc, if you are given any of these kinds of surfaces and asked to measure the distance interval between any two points on the given surface. Generally, you pick up the standard 10 Rs ruler and mark two points on the surface, having the required distance, and just measure it. This action is suitable only if the surface is plane as well and the edge of the ruler is uniformly marked (calibrated). In the case of surfaces with valleys and peaks, using such a standard ruler may cause you may fall into difficulty. For resolving, you can either use a wool thread and measure the required distance in two steps, or you can entirely use a different ruler built uniquely for the surface with valleys and peaks. Physicists do both things; instead of using wool thread, they invoke transformation, and in the case of using a unique ruler, they use a metric. Since Minkowski spacetime is a hyperbolic surface, as a result we can not use a Euclidean ruler and should apply a hyperbolic ruler instead.

The name of this hyperbolic ruler, if the surface is also flat, is the Minkowski metric (having no mass placed on the surface) and the Schwarzschild Metric (having mass); otherwise, in the case of curved spacetime, we get the Kerr metric, etc. One of the conclusions should bear in mind that the metric captures all the geometric and causal structures of the given surface. A few examples of (mostly plus) metrics:

- 1. Euclidean metric in both Spherical coordinates and Cartesian coordinates: Flat space

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2 \theta \end{bmatrix} \equiv \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- 2. Minkowski metric in Spherical coordinates: Flat spacetime, Spherical and empty

$$\eta_{\mu\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{bmatrix}$$

- 3. The Schwarzschild Metric in Spherical coordinates: Conformally flat (uncharged, non-rotating mass or static; $r_s = \frac{2GM}{c^2}$ which in natural units($G = c = 1$) gives $r_s = 2M$)

$$g_{\mu\nu} = \begin{bmatrix} -1 \left(1 - \frac{2M}{r}\right) & 0 & 0 & 0 \\ 0 & \frac{1}{\left(1 - \frac{2M}{r}\right)} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{bmatrix}$$

- 4. Kerr metric: uncharged, rotating mass, ellipsoid

$$g_{\mu\nu} = \begin{bmatrix} -\left(\frac{r^2 - 2M + a^2}{\rho^2}\right) & 0 & 0 & 0 \\ 0 & \frac{\rho^2}{r^2 - 2M + a^2} & 0 & 0 \\ 0 & 0 & \rho^2 & 0 \\ 0 & 0 & 0 & \frac{(r^2 + a^2)^2 \sin^2 \theta}{\rho^2} \end{bmatrix}$$

- 5. Reissner–Nordström metric: A charged, non-rotating mass
- 6. Kerr–Newman metric: A charged, rotating mass
- 7. Kerr–Newman–de Sitter metric: Electrically charged, rotating mass embedded in an expanding universe.

Some uses of metric apart from those stated above:

- If x^ν is a given four-distance, then we can get $x_\mu = \eta_{\mu\nu} x^\nu$. Where $\eta_{\mu\nu}$ is a metric.
- In case of taking the Lorentzian inner product: $A \cdot B = A^\mu B_\mu = \eta_{\mu\nu} A^\mu B^\nu$. Thus, we see we can find the product even if we have the same type of two different four vectors.

3.2 Recall Mathematicians

3.2.1 Covariant derivative

Just check equation (4.1). If you noticed the sequence of our desired goals, you would have found that they involve an incremental degree of mathematical sophistication, to achieve the fact that all is due to the curvature of the spacetime manifold.

Let's start then,

If I tell you that the derivative is incapable at its core when one tries to tackle the problem of curvature and symmetries of spacetime. Suppose you have a trajectory designated by an affine parameter λ . Let Piyush be at point P carrying an arrow U_P , and Qadir is at point Q on the trajectory, also has his own U_Q . But, both of them are $\Delta\lambda$ apart, sadly.

We define the (slope) derivative at a point P as,

$$\frac{dU}{d\lambda} = \lim_{\Delta\lambda \rightarrow 0} \frac{U_Q - U_P}{\Delta\lambda}$$

Now, ask, do we really know how to subtract these two arrows (vectors) about a point P, provided we are keeping Piyush and Qadir *standstill* all along the process of determining the slope about point P? The answer is straight -NO. To apply the vector addition rule, their tails should meet at least. Therefore, the only way to cure the issue is to let Qadir move from his point Q to Piyush's Point P. In text, we call this *transportation*. Here, we will see the transportation of two different kinds: Parallel transport and Lie transport. Each mode of transportation sets a set of logical constraints on the passenger (Qadir or Piyush), by what pieces of information one is aiming for. By Parallel transport, our aim is to get realised of potential curvature, and by Lie transport, we investigate the potential presence of symmetries hidden in spacetime, when moved along a certain way; These things will be cleared up soon. Additionally and importantly, in parallel transport we keep maintaining the initial state of the moving arrow, but in Lie transport we have to take care of relative orientation of arrow with respect to path, thus twist and turns are allowed steps, but angle with respect to path should be kept fixed, thus we let the tail of arrow to constantly slid along path untill it reaches the required end point, about which the derivative (slope) is instructed to be determined.

Let denote transportation from Q to P as $U_{Q \rightarrow P}$. As a vector $\vec{a}$ in component-basis form is expressed as $a^\mu \frac{\partial}{\partial x^\mu}$, analogously, we have $(U^\mu \frac{\partial}{\partial x^\mu})_{Q \rightarrow P}$. Also, don't forget that U_Q is just $U_{(P+\Delta\lambda)}$. Say λ_0 is location of P , now Taylor expansion of component of U_Q ,

$$\begin{aligned} U_Q^\mu &= U^\mu(\lambda_0 + \Delta\lambda) = U^\mu(\lambda_0) + \frac{\partial U^\mu}{\partial \lambda}|_{\lambda_0} \Delta\lambda + \dots \\ &\approx U_P^\mu + \frac{\partial U^\mu}{\partial \lambda}|_P \Delta\lambda \end{aligned}$$

Ain't we need to deal with $U_{Q \rightarrow P}^\mu$ instead of U_Q^μ ? Yes. Thus, before this approximated Taylor expansion, we should mathematically represent the transportation, and we can achieve this by a change of basis

$$\frac{\partial}{\partial x^\mu}|_{Q \rightarrow P} = C_\mu^\nu \frac{\partial}{\partial x^\nu}|_P$$

Here, C_μ^ν represents the magnitude of change in the basis vector when parallel transport is made.

$$\frac{dU}{d\lambda} = \lim_{\Delta\lambda \rightarrow 0} \left(\frac{U_Q^\mu C_\mu^\nu - U_P^\nu}{\Delta\lambda} \right) \frac{\partial}{\partial x^\nu} \Big|_P$$

Note expression in the bracket is just the component of the derivative of vector U , So,

$$\left(\frac{dU}{d\lambda} \right)^\nu = \lim_{\Delta\lambda \rightarrow 0} \left(\frac{U_Q^\mu C_\mu^\nu - U_P^\nu}{\Delta\lambda} \right)$$

Now, inserting the Taylor series expansion, the derivative has the form,

$$\left(\frac{dU}{d\lambda} \right)^\nu = \lim_{\Delta\lambda \rightarrow 0} \left(\frac{U_Q^\mu C_\mu^\nu - U_P^\nu}{\Delta\lambda} \right) \approx \lim_{\Delta\lambda \rightarrow 0} \left(\frac{(U_P^\mu + \frac{\partial U^\mu}{\partial \lambda} \Big|_P \Delta\lambda) C_\mu^\nu - U_P^\nu}{\Delta\lambda} \right)$$

At this stage, we get three tasks: 1. Define: $U_P^\nu = \delta_\mu^\nu U_P^\mu$, 2. define $\lambda = x^\alpha$ and 3. open the inner small parenthesis. We left with,

$$\left(\frac{dU}{dx^\alpha} \right)^\nu = \frac{\partial U^\nu}{\partial x^\alpha} \Big|_P + \lim_{\Delta x^\alpha \rightarrow 0} \left(\frac{C_\mu^\nu - \delta_\mu^\nu}{\Delta x^\alpha} \right) U_P^\mu$$

You might be wondering why the first term should instead be $\frac{\partial U^\mu}{\partial x^\alpha} \Big|_P C_\mu^\nu$. Note and remember that, we further approximated by taking $C_\mu^\nu = \delta_\mu^\nu + \mathcal{O}(\Delta\lambda)$, which upon contracting (ignoring higher order) gives $U_{,\alpha}^\nu$ only.

The fourth task, which I now mention, is to define this object $C_{\mu\alpha}^\nu = \lim_{\Delta x^\alpha \rightarrow 0} \left(\frac{C_\mu^\nu - \delta_\mu^\nu}{\Delta x^\alpha} \right)$.

We quickly recognise this as an affine connection. So if this affine connection also respects “the condition” (it will soon be disclosed before you), then we replace it with Christoffel connection, $\Gamma_{\mu\alpha}^\nu$.

Finally, the covariant derivative of the vector component U^μ is,

$$\nabla_\alpha U^\mu = U_{,\alpha}^\mu + C_{\mu\alpha}^\nu U^\mu$$

One should remember that we can generalise (n,m) tensors; for example, here this is for a mixed tensor (2,2), as

$$\nabla_\mu T_{\gamma\delta}^{\alpha\beta} = T_{\gamma\delta,\mu}^{\alpha\beta} + C_{\mu\lambda}^\alpha T_{\gamma\delta}^{\lambda\beta} + C_{\mu\lambda}^\beta T_{\gamma\delta}^{\alpha\lambda} - C_{\mu\gamma}^\lambda T_{\lambda\delta}^{\alpha\beta} - C_{\mu\delta}^\lambda T_{\gamma\lambda}^{\alpha\beta}$$

For usefulness, one should remember the action of covariant derivative on metric tensor (rank 2)[this is ‘the condition’ we were talking about],

$$\begin{aligned} \nabla_\mu g_{\alpha\beta} &= 0 \\ \implies g_{\alpha\beta,\mu} &= \Gamma_{\mu\beta}^\lambda g_{\lambda\alpha} + \Gamma_{\mu\alpha}^\lambda g_{\lambda\beta} \end{aligned}$$

And how can one rewrite the geodesic equation? Just a few pages back, we derived the Geodesic equation (aka Newton’s 2nd law LHS expression),

Now, since we are probably aware of the ABC of covariant derivative, can we try to compactly rewrite the Geodesic equation? (consider τ is being time of particles in their own clock) Let's try,

$$\begin{aligned} \frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} &= 0 \\ \implies \frac{d}{d\tau} \left(\frac{dx^\mu}{d\tau} \right) + \Gamma_{\alpha\beta}^\mu \dot{x}^\alpha \dot{x}^\beta &= 0 \end{aligned}$$

Since you have a better eye than me and hence, you will notice that expression in small brackets as a vector component, say V^μ .

$$\implies \frac{d}{d\tau} \dot{V}^\mu + \Gamma V^\alpha V^\beta = 0$$

And, completely first term can be seen as the rate of change of the tangent vector along the curve, parametrised by $x^\mu(\tau)$.

Recalling the covariant derivative of the vector component (V^α) [In the context of spacetime],

$$\nabla_\mu V^\alpha = \partial_\mu V^\alpha + \Gamma_{\mu\lambda}^\alpha V^\lambda$$

Multiply by V^μ ,

$$V^\mu \nabla_\mu V^\alpha = V^\mu \partial_\mu V^\alpha + \Gamma_{\mu\lambda}^\alpha V^\lambda V^\mu$$

Then, taking help from your friend, you show the first term on the RHS,

$$V^\mu \partial_\mu V^\alpha = \frac{dx^\mu}{d\tau} \left(\frac{\partial}{\partial x^\mu} \right) \frac{dx^\alpha}{d\tau} = \frac{\partial}{\partial x^\mu} \left(\frac{dx^\mu}{d\tau} \right) \frac{dx^\alpha}{d\tau} = \frac{d}{d\tau} \left(\frac{dx^\alpha}{d\tau} \right) = \frac{d^2 x^\alpha}{d\tau^2}$$

Since we can consider $V^\mu = \dot{x}^\mu$ and $V^\lambda = \dot{x}^\lambda$.

On comparing the Geodesic equation with the above expression, we see, if the vector components $V^\lambda \equiv \dot{x}^\lambda$ and $V^\mu \equiv \dot{x}^\mu = \frac{\partial}{\partial \tau} x^\mu$, give,

$$\implies V^\mu \partial_\mu V^\alpha = \ddot{x}^\alpha + \Gamma_{\mu\lambda}^\alpha \dot{x}^\mu \dot{x}^\lambda = 0$$

Thus, we rewrite the Geodesic equation as,

$$V^\mu \nabla_\mu V^\alpha = 0$$

It just expresses the fact that parallel transport along the same minimal length is zero. Now, let's list the situations where some of these results yield zero:

- Suppose Piyush and Qadir are on a straight path, and their arrows are mutually parallel to each other. (Why is this always true? These arrows are part of a common vector field.) Now, whenever Qadir performs parallel transport to point P and we then determine the covariant derivative, the “subtraction part” of the arrow (i.e., the magnitude of the covariant derivative of the vector component at the fixed point P) will intuitively be zero.
- Suppose both persons keep their arrows tangential to one another. Then, when the tangential arrow reaches a point, we do not notice any deviation (since a tangent does not change along its direction, parallel transported). So again, there is no “subtraction part,” and thus the magnitude of the covariant derivative is zero.

You might be wondering how these visualisations are useful. For me, the second point highlights the reason why $\nabla_\alpha g_{\mu\nu} = 0$, while the first point shows that parallel transport indeed senses whether a trajectory is curved or not. This trajectory can be extended to a manifold, and using the same technique, we can gain insight into whether the manifold is flat or curved.

A mathematician speaks. A metric allows the measurement of a line element if provided with two input vectors. Let the Length of vector V be $g(V, V)$, you see that metric g indeed requires two inputs. If these are different, then $g(V, W)$.

Taking differentiation along the line x^α , we see,

$$g_{,\alpha}(V, W) = (g_{\mu\nu} V^\mu W^\nu)_{,\alpha} = g_{\mu\nu,\alpha} V^\mu W^\nu + g_{\mu\nu} V^\mu_{,\alpha} W^\nu + g_{\mu\nu} V^\mu W^\nu_{,\alpha}$$

What's next now, after using Leibniz's rule above? Note that we make a few considerations about V and W ; we call their covariant derivative zero. (Why?- That was the condition- we are interested in spacetime (or, Parallel transport along the same minimal length is zero:), and our affine connection should be replaced by $\Gamma^\nu_{\mu\alpha}$).

Then, a respective covariant derivative of their vector components is

$$\begin{aligned} \nabla_\alpha V^\mu &= V^\mu_{,\alpha} + \Gamma^\mu_{\nu\alpha} V^\nu = 0 \implies V^\mu_{,\alpha} = -\Gamma^\mu_{\nu\alpha} V^\nu \\ \nabla_\alpha W^\nu &= W^\nu_{,\alpha} + \Gamma^\nu_{\mu\alpha} W^\mu = 0 \implies W^\nu_{,\alpha} = -\Gamma^\nu_{\mu\alpha} W^\mu \end{aligned}$$

This also says that the inner product indicates that $g_{,\alpha}(V, W) = 0$ as well.

Substitute these,

$$g_{,\alpha}(V, W) = (g_{\mu\nu} V^\mu W^\nu)_{,\alpha} = g_{\mu\nu,\alpha} V^\mu W^\nu - g_{\mu\nu} \Gamma^\nu_{\mu\alpha} V^\mu W^\nu - g_{\mu\nu} V^\mu \Gamma^\nu_{\mu\alpha} W^\mu = 0$$

You see above expression is wrong unless we correctly reset the indices,

$$= (g_{\mu\nu,\alpha} - g_{\beta\nu} \Gamma^\beta_{\mu\alpha} - g_{\mu\beta} \Gamma^\beta_{\nu\alpha}) V^\mu W^\nu = 0$$

Since, $V^\mu W^\nu \neq 0$, Therefore,

$$g_{\mu\nu,\alpha} - g_{\beta\nu} \Gamma^\beta_{\mu\alpha} - g_{\mu\beta} \Gamma^\beta_{\nu\alpha} = 0$$

Now let make two more copies of this expression $(\alpha, \mu, \nu) \rightarrow (\mu, \nu, \alpha)$ and $(\mu, \nu, \alpha) \rightarrow (\nu, \alpha, \mu)$, and then adding the first two and subtraing the last,

We get,

$$g_{\mu\nu,\alpha} - g_{\nu\alpha,\mu} - g_{\alpha\mu,\nu} + g_{\alpha\beta} \Gamma^\beta_{\mu\nu} = 0$$

Let $\beta \rightarrow \lambda$, or, using $\Gamma^\lambda_{\mu\nu} = g_{\alpha\beta} \Gamma^\beta_{\mu\nu} g^{\alpha\lambda}$

$$\implies \Gamma^\lambda_{\mu\nu} = \frac{1}{2} g^{\lambda\alpha} (-g_{\mu\nu,\alpha} + g_{\nu\alpha,\mu} + g_{\mu\alpha,\nu})$$

Henceforth, we have derived the expression of the Christoffel symbol in terms of the metric.

Let's take a digression and, in particular, analyse the Christoffel symbol. As it is a three-indexed object, we estimate the total number of equations it conveys, $4^3 = 64$, out of which independent equations are 40. Now consider the following,

$$64 : \text{Total} \equiv 40 : \text{symmetric} + 24 : \text{anti-symmetric}$$

$$\Gamma_{\mu\nu}^{\lambda} = \frac{1}{2} (\Gamma_{\mu\nu}^{\lambda} + \Gamma_{\nu\mu}^{\lambda}) + \frac{1}{2} (\Gamma_{\mu\nu}^{\lambda} - \Gamma_{\nu\mu}^{\lambda})$$

Here, The second term is regarded as Torsion, $T_{\mu\nu}^{\lambda} = \frac{1}{2} (\Gamma_{\mu\nu}^{\lambda} - \Gamma_{\nu\mu}^{\lambda})$

It is an assumption that, in the study of general relativity, the torsion part vanishes.

Thus, it allows us to consider that the Christoffel symbol is symmetric under the exchange of lower indices. Based on this assumption about Torsion, we derive the explicit form of the Christoffel symbol in terms of the metric.

Riemannian curvature

I told you that parallel transport, which links to the covariant derivative, somehow captures curvature. You would be a bit stressed to know that, explicitly, curvature is related to the Riemann curvature tensor. Which involves a bit more sophisticated form. Still, at the heart of all these lies metric and its higher-order differentials.

Is there something special we are going to do? I must say that not something so special. First of all, learn that there are two kinds of curvature, namely, intrinsic curvature and extrinsic curvature (Whose determination depends on embedding in higher higher-dimensional space). Since our interest is not only codifying the presence of potential curvature in spacetime, but also we lack any sources which hint that spacetime itself is embedded in higher stuff, we investigate the intricacies of intrinsic curvature only. For example, an A4-size paper is both intrinsically and extrinsically flat, and if one bends that paper, or part of a thick notebook, it indeed remains intrinsically flat but gets extrinsically curved.

To check curvature, we consider a manifold, then we do a parallel transport of a vector over an *infinitesimal*-sized loop (why?) parametrised by some parameter. If we say the inner product of the transported vector with the local co-vector remains intact with respect initial configuration, then no curvature (really?); otherwise, we detect the desired result. For example, the sum of angles of a triangle drawn on a flat surface adds to π , but if the surface is curved (on a sphere), we do not expect π . Just like we heard about intrinsic curvature and extrinsic curvature, we also have two other kinds, positive and negative curvature [In case of a triangle, positive if: $\sum \theta > \pi$ and negative if: $\sum \theta < \pi$]. So it may be a possibility that the final configuration appears intact at the end, just due to positive and negative curvature equally cancelling each other's contribution. So, no change in configuration does not directly map to the idea that the manifold has finite non-zero curvature. That's the reason we stressed, parallel transport of a vector over an *infinitesimal*-sized loop.

Now suppose you stick a coordinate axis on the infinitesimal loop, marked by two fixed axes, having components T^{μ} and S^{μ} , such that the parallel transported vector component V^{α} satisfies $T^{\mu} \nabla_{\mu} V^{\alpha} = 0$, and $S^{\mu} \nabla_{\mu} V^{\alpha} = 0$; these conditions are referred to as coordinate compatibility (Geodesic equations). This simply translates to, parallel transport of V^{α} along S^{μ} and T^{μ} is like moving along its geodesics. In addition, we recall the metric-compatibility condition $\nabla_{\mu} (g_{\alpha\beta}) = 0$.

We talked about all the required ingredients, but now ask which is the observable (measurable), we always need to look for whether 'it' is intact or not? You may have guessed rightly, the inner product $V^{\mu} W_{\mu}$ is the quantity of interest. Once we complete the one-way parallel transportation of V^{μ} (while comparing any deviation from local W_{μ} at the initial and final stage), we will repeat the process to determine $W^{\mu} V_{\mu}$, and again compare the initial and final stage inner product; because we are not sure these are commutative or not, (unless we impose). For those who are still guessing, why is our quantity of interest an inner product?- because it is a scalar quantity, we know the action of the covariant derivative on a scalar. Just to keep things simple, let's consider an infinitesimal

loop (square-like), having T^μ and S^μ , forming corners $(0,0)$, $(\Delta t, 0)$, $(\Delta t, \Delta s)$ and $(0, \Delta s)$, Where, Δt is infinitesimal separation of S^μ at $(0,0)$ and S^μ at $(\Delta t, 0)$, similarly, Where, Δs is infinitesimal separation of T^μ at $(0,0)$ and T^μ at $(0, \Delta s)$.

Now find,

$$W^\mu V_\mu|_{(\Delta t, 0)} - W^\mu V_\mu|_{(0, 0)} = f(0 + \Delta t, 0) - f(0, 0) \approx \Delta t \frac{\partial f}{\partial t}|_{(0, 0)}$$

Using mid-point approximation, result be,

$$\delta(W^\mu V_\mu)_{(0, 0) \rightarrow (\Delta t, 0)} \approx \Delta t \frac{\partial f}{\partial t}|_{(\frac{\Delta t}{2}, 0)} = \Delta t T^\mu \nabla_\mu (W^\alpha V_\alpha)|_{(\frac{\Delta t}{2}, 0)}$$

This result in general coordinate invariance takes the form for a full-loop (Hints: 4 such terms):

$$\begin{aligned} \delta(W^\mu V_\mu)_{\text{full-loop}} &= \Delta t T^\mu \nabla_\mu (W^\alpha V_\alpha)|_{(\frac{\Delta t}{2}, 0)} + \Delta s S^\mu \nabla_\mu (W^\alpha V_\alpha)|_{(\Delta t, \frac{\Delta s}{2})} \\ &\quad - \Delta t T^\mu \nabla_\mu (W^\alpha V_\alpha)|_{(\frac{\Delta t}{2}, \Delta s)} - \Delta s S^\mu \nabla_\mu (W^\alpha V_\alpha)|_{(0, \frac{\Delta s}{2})} \end{aligned}$$

Now, from above divide the transportation into two sets as follows: $(\frac{\Delta t}{2}, 0) \rightarrow (\frac{\Delta t}{2}, \Delta s)$, and $(0, \frac{\Delta s}{2}) \rightarrow (\Delta t, \frac{\Delta s}{2})$

Which respectively gives,

$$\begin{aligned} \delta(W^\mu V_\mu)_{(\frac{\Delta t}{2}, 0) \rightarrow (\frac{\Delta t}{2}, \Delta s)} &= \Delta t \left[T^\mu \nabla_\mu (W^\alpha V_\alpha)|_{(\frac{\Delta t}{2}, 0)} - T^\mu \nabla_\mu (W^\alpha V_\alpha)|_{(\frac{\Delta t}{2}, \Delta s)} \right] \\ &= -\Delta t \Delta s S^\nu [V_\alpha T^\mu \nabla_\nu (\nabla_\mu W^\alpha) + \nabla_\nu (V_\alpha T^\mu) \nabla_\mu W^\alpha] \end{aligned}$$

And,

$$\begin{aligned} \delta(W^\mu V_\mu)_{(0, \frac{\Delta s}{2}) \rightarrow (\Delta t, \frac{\Delta s}{2})} &= \Delta s \left[S^\mu \nabla_\mu (W^\alpha V_\alpha)|_{(\Delta t, \frac{\Delta s}{2})} - S^\mu \nabla_\mu (W^\alpha V_\alpha)|_{(0, \frac{\Delta s}{2})} \right] \\ &= \Delta t \Delta s T^\nu [V_\alpha S^\mu \nabla_\nu (\nabla_\mu W^\alpha) + \nabla_\nu (V_\alpha S^\mu) \nabla_\mu W^\alpha] \end{aligned}$$

From adding these two,

We finally get,

$$\begin{aligned} \delta(W^\mu V_\mu)_{\text{full-loop}} &= \Delta t \Delta s V_\alpha [-S^\nu T^\mu \nabla_\nu \nabla_\mu W^\alpha + S^\mu T^\nu \nabla_\nu \nabla_\mu W^\alpha] \\ &\quad + \Delta t \Delta s (\nabla_\mu W^\alpha) [-S^\nu \nabla_\nu (V_\alpha T^\mu) + T^\mu \nabla_\nu (V_\alpha S^\mu)] \end{aligned}$$

Which, upon simplification,

$$\delta(W^\mu V_\mu)_{\text{full-loop}} = \Delta t \Delta s T^\mu S^\nu V_\alpha [\nabla_\mu, \nabla_\nu] W^\alpha$$

Note: $[\nabla_\mu, \nabla_\nu]$ is Lie bracket. Thus, if this Lie bracket vanishes, then the surface is said to be intrinsically flat. It will capture the presence of any local curvature in spacetime.

Say, if the presence of curvature is a general coordinate invariant concept, then,

$$[\nabla_\mu, \nabla_\nu] W^\alpha$$

must be a tensor.

So, We define, Riemann curvature tensor, $R_{\alpha\mu\nu}^\beta$ as,

$$[\nabla_\mu, \nabla_\nu]W_\alpha = R_{\alpha\mu\nu}^\beta W_\beta$$

By the formulation of the covariant derivative and its action, one can get the explicit form of $R_{\alpha\mu\nu}^\beta$ in terms of the metric or its higher-order derivative.

We indeed find,

$$[\nabla_\mu, \nabla_\nu]W_\alpha = (\partial_\nu \Gamma_{\mu\alpha}^\rho - \partial_\mu \Gamma_{\nu\alpha}^\rho + \Gamma_{\mu\alpha}^\lambda \Gamma_{\nu\lambda}^\rho - \Gamma_{\nu\alpha}^\lambda \Gamma_{\mu\lambda}^\rho)W_\rho$$

Thus,

$$R_{\alpha\mu\nu}^\rho = \partial_\nu \Gamma_{\mu\alpha}^\rho - \partial_\mu \Gamma_{\nu\alpha}^\rho + \Gamma_{\mu\alpha}^\lambda \Gamma_{\nu\lambda}^\rho - \Gamma_{\nu\alpha}^\lambda \Gamma_{\mu\lambda}^\rho$$

Now, we look at some properties of this curvature tensor.

- Upon $\mu \leftrightarrow \nu$, one shows that $R_{\alpha[\mu\nu]}^\rho$ is anti-symmetric in bottom two indices:

$$R_{\alpha\mu\nu}^\rho = -R_{\alpha\nu\mu}^\rho$$

- Next, upon $\rho \leftrightarrow \alpha$: No new pattern seen.
- How to lower the upper index?, to achieve, as always we invoke the metric $g_{\mu\nu}$.

$$R_{\mu\nu\sigma\rho} = g_{\rho\lambda} R_{\mu\nu\sigma}^\lambda = R_{\sigma\rho\mu\nu} = -R_{\rho\nu\sigma\mu} = -R_{\nu\mu\sigma\rho} = -R_{\mu\nu\rho\sigma}$$

And its explicit form is,

$$R_{\mu\nu\sigma\rho} = \frac{1}{2} (g_{\mu\rho, \nu\sigma} + g_{\nu\sigma, \mu\rho} - g_{\mu\sigma, \nu\rho} - g_{\nu\rho, \mu\sigma}) + g_{\alpha\beta} (\Gamma_{\mu\sigma}^\alpha \Gamma_{\nu\rho}^\beta - \Gamma_{\mu\rho}^\alpha \Gamma_{\nu\sigma}^\beta)$$

- $R_{\mu\nu\alpha\beta} g^{\mu\nu} g^{\alpha\beta} = 0$ always holds. As $R_{\mu\nu\alpha\beta}$ is anti-symmetric in the first two indices and the last two indices, whereas g^{ab} is symmetric.
- In local inertial frame(LIF):

$$R_{\alpha\beta\mu\nu} \stackrel{LIF}{=} \frac{1}{2} (g_{\alpha\nu, \beta\mu} - g_{\alpha\mu, \beta\nu} + g_{\beta\mu, \alpha\nu} - g_{\beta\nu, \alpha\mu})$$

Because,

$$\Gamma_{\mu\nu, \sigma}^\alpha \stackrel{LIF}{=} \frac{1}{2} g^{\alpha\beta} (g_{\beta\mu, \nu\sigma} + g_{\beta\nu, \mu\sigma} - g_{\mu\nu, \beta\sigma})$$

- Few named tensors and scalars are:

Reimann Tensor (0,4) : $R_{\alpha\beta\mu\nu}$

Kreischmann Scalar (0,0) : $R_{\alpha\beta\mu\nu} R^{\alpha\beta\mu\nu}$

Ricci Tensor (0,2) : $R_{\beta\nu} = R_{\alpha\beta\mu\nu} g^{\alpha\mu}$

Ricci Scalar (0,0) : $R = R_{\beta\nu} g^{\beta\nu}$

Reimann Scalar (0,0) : $R_{\beta\nu} R^{\beta\nu}$

3.2.2 Lie and Killing stuff

Sometimes in life, one cannot multitask. In my case, I was thinking of simultaneously discussing parallel and Lie transport, but you see how bringing the discussion of curvature amidst all has ruined my desire. Anyway, Lie transport is not the generalisation of parallel transport. Instead, the existence of such transport motivates us to ponder more about such ways of transport apart from the known ones.

This time, we will let Piyush move to point Q from Point P. This time, Piyush does not need to worry about maintaining parallel state of his arrow (with respect to initial state) when he moves, rather he should feel the texture of natural flow, or trajectory (integral curves), by which I mean, that he can turn or twist direction of arrow (this breaks the parallelism) in accordance to path he is travelling on, or just to match the path's curvature. For instance, if the path takes a sharp left turn, his arrow should be adjusted such that the arrow is kept at exact *orientation* or fixed relative angle with respect to the path, as it was before (here, turning the arrow anticlockwise). In short, if the initial state makes an angle θ , then along every point during transportation, this angle should be kept fixed. You see, I am not a good writer, so you should be sketching vector transport. If I remind you about parallel transport, it is like keeping the vector as straight as possible relative to its initial direction, despite the path's curvature and in Lie, we respect the integral flow curve. Now, you might have noticed the difference.

Also, I hope you might know the meaning of the following expression,

$$\frac{dV}{d\lambda} = \lim_{\Delta\lambda \rightarrow 0} \frac{V_Q - V_{P \rightarrow Q}}{\Delta\lambda}$$

A question arises about how to mathematically (precisely) incorporate the meaning of 'Lie' in this transport. So to do that, we cover the manifold (via parallel transport) by several copies of the main trajectory, and consider that Piyush and Qadir are to be present on just one of the integral curves [main trajectory], it seems like a boat in a streamline of the flowing river (set of copied trajectories). Now, these integral curves are made possible via parallel transporting of the tangential vector $[t = \frac{\partial}{\partial\lambda}]$, thus its component is $t^\mu = x^\mu_{,\lambda}$ of Piyush's trajectory $[x^\mu(\lambda)]$. Next thing we do is to define that the arrows are part of the integral curve (at P) as $x^\mu(\gamma)$, so let's say on the integral curve $x^\mu(\gamma)$ lies a Point R , separated by PR displacement vector $\Delta\gamma V_P$ from point P and we let this to flow (Lie transport), and suppose its tail reached the point Q , by then its head touched point S , which is on integral curve passing through R and parallel along PQ separated by amount $\Delta\lambda t_R$ from R and $\Delta\lambda t_P$ from P , then via vector law, we identify displacement vector QS , as $\Delta\gamma V_{P \rightarrow Q}$, then the subtraction (displacement vector $\Delta\lambda t^\mu = \Delta x^\mu = x^\mu_Q - x^\mu_P$) represents the Lie derivative,

under this discussed setting, recalling the above expression and if multiplying by $\Delta\gamma$,

$$\Delta\gamma \frac{dV}{d\lambda} = \lim_{\Delta\lambda \rightarrow 0} \frac{\Delta\gamma V_Q - \Delta\gamma V_{P \rightarrow Q}}{\Delta\lambda}$$

Now we write the corresponding displacement vector and do the Taylor expansion,

- Component of displacement vector (Lie transport) from point P to Q:

$$\begin{aligned} \Delta\gamma V_{P \rightarrow Q}^\mu &= (x_P^\mu + \Delta\gamma V_P^\mu + \Delta\lambda t_R^\mu) - (x_P^\mu + \Delta\lambda t_P^\mu) \\ &= \Delta\gamma V_P^\mu + \Delta\lambda t_R^\mu - \Delta\lambda t_P^\mu \end{aligned}$$

And, Taylor expansion of V_Q^μ ,

$$V_Q^\mu = V^\mu(x_Q^\alpha) = V^\mu(x_P^\alpha + \Delta\lambda t_P^\alpha) \approx V^\mu(x_P^\alpha) + \Delta\lambda t_P^\alpha V_{,\alpha}^\mu$$

- Component of displacement vector $[\Delta\lambda t_P$, at point P] from point P to Q (Lie transport), that is just the subtraction of coordinates of point Q and coordinates of point P .

$$\Delta\lambda t_{P \rightarrow Q}^\mu = x_Q^\mu - x_P^\mu$$

And, Taylor expansion of t_R^μ ,

$$t_R^\mu = t^\mu(x_R^\alpha) = t^\mu(x_P^\alpha + \Delta\gamma V_P^\alpha) \approx t^\mu(x_P^\alpha) + \Delta\gamma V_P^\alpha t_{,\alpha}^\mu$$

Note the roll is switched as $t \leftrightarrow V$, $\lambda \leftrightarrow \gamma$.

then,

$$\Delta\gamma \left(\frac{dV}{d\lambda} \right)^\mu = \lim_{\Delta\lambda \rightarrow 0} \frac{1}{\Delta\lambda} [\Delta\gamma \Delta\lambda t_P^\alpha V_{,\alpha}^\mu - \Delta\lambda \Delta\gamma V_P^\alpha t_{,\alpha}^\mu]$$

Thus, component of Lie derivative of vector V with respect to tangential vector t be,

$$\left(\frac{dV}{d\lambda} \right)^\mu = [t_P^\alpha V_{,\alpha}^\mu - V_P^\alpha t_{,\alpha}^\mu] = \mathcal{L}_t V^\mu$$

Now we will list some important results:

1. Action of Lie derivative on scalar:

$$\mathcal{L}_t f = t^\alpha f_{,\alpha}$$

2. Action of Lie derivative on covector: If this $f = W_\mu V^\mu$, then,

$$\mathcal{L}_t(W_\mu V^\mu) = (\mathcal{L}_t W_\mu) V^\mu + W^\mu (\mathcal{L}_t V_\mu) = (\mathcal{L}_t W_\mu) V^\mu + W^\mu t^\alpha V_{,\alpha}^\mu - W^\mu V^\alpha t_{,\alpha}^\mu$$

Also note, as $f = W_\mu V^\mu \implies \mathcal{L}_t f = t^\alpha (W_\mu V^\mu)_{,\alpha} \stackrel{\text{Product rule}}{=} t^\alpha (W_\mu)_{,\alpha} V^\mu + t^\alpha W_\mu (V^\mu)_{,\alpha}$
Comparing both of these, we end up having (after setting $V^\mu = 1$),

$$\mathcal{L}_t W_\mu = t^\alpha W_{\mu,\alpha} + W_\alpha t_{,\mu}^\alpha$$

3. On metric:

$$\mathcal{L}_t g_{\mu\nu} = t^\alpha g_{\mu\nu,\alpha} + g_{\alpha\nu} t_{,\mu}^\alpha + g_{\mu\alpha} t_{,\nu}^\alpha$$

4. On mixed tensor (n, m) :

$$\begin{aligned} \mathcal{L}_t T_{\nu_1 \nu_2 \dots \nu_m}^{\mu_1 \mu_2 \dots \mu_n} &= t^\alpha T_{\nu_1 \nu_2 \dots \nu_m, \alpha}^{\mu_1 \mu_2 \dots \mu_n} - [T_{\nu_1 \nu_2 \dots \nu_m}^{\alpha \mu_2 \dots \mu_n} t_{,\alpha}^{\mu_1} + T_{\nu_1 \nu_2 \dots \nu_m}^{\mu_1 \alpha \dots \mu_n} t_{,\alpha}^{\mu_2} + \dots] \\ &\quad + [T_{\alpha \nu_2 \dots \nu_m}^{\mu_1 \mu_2 \dots \mu_n} t_{,\nu_1}^\alpha + T_{\nu_1 \alpha \dots \nu_m}^{\mu_1 \mu_2 \dots \mu_n} t_{,\nu_2}^\alpha + \dots] \end{aligned}$$

We know that metric is get expressed in terms of spacetime coordinates, potentially in multitude of coordinate system, and suppose for a certain metric you notice that some of the coordintes or their combinations are absent; this points out that the metric is independent of such one of the coordinats or their combinations, say x^0 , then $\frac{\partial}{\partial x^0}$ is regarded as killing vector field (KVF). Also, we perceive that the spacetime is symmetric under translation along the x^0 coordinate lines (integral curve).

We know that if spacetime is symmetric along the integral curve of a vector field K , if

$$\mathcal{L}_K g_{\mu\nu} = 0$$

Then, K is regarded as a Killing vector field (KVF). We designate the components of KVF as K^α .

From the expression of the Lie derivative of the metric, we see,

$$\mathcal{L}_K g_{\mu\nu} = K^\alpha \partial_\alpha g_{\mu\nu} + g_{\alpha\nu} \partial_\mu K^\alpha + g_{\mu\alpha} \partial_\nu K^\alpha = 0$$

I will not mention the proof, but one can replace partial derivatives by covariant derivatives in the above equation, then,

$$\mathcal{L}_K g_{\mu\nu} = g_{\alpha\nu} \nabla_\mu K^\alpha + g_{\mu\alpha} \nabla_\nu K^\alpha = 0$$

Here, we used the fact $\nabla_\alpha g_{\mu\nu} = 0$

After contractions, we find the condition,

$$\mathcal{L}_K g_{\mu\nu} \equiv \nabla_\mu K_\nu + \nabla_\nu K_\mu = 0$$

For example, In the Schwarzschild spacetime given by the line element

$$ds^2 = -c^2 \left(1 - \frac{2GM}{c^2 r}\right) dt^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$$

One can easily notice that the metric is independent of the time coordinate (t) and azimuthal angle (ϕ). So without further ado, we say that components of the Killing vector field are $K^0 = \frac{\partial}{\partial t}$ and $K^3 = \frac{\partial}{\partial \phi}$; Which suggest that the mertic should be both symmetric about time and azimuthal angle, that physically this corresponds to metric being static and spherically symmetric. Under a spherical coordinate system, for a given case, we represent the Killing vector field $K = (1, 0, 0, 1)$.

3.2.3 Final touch on metric

Under the discussion on Riemann curvature tensor, we note that a Ricci Tensor is being,

$$R_{\mu\nu} = R_{\mu\alpha\nu\beta} g^{\alpha\beta}$$

It is essential to mention that $R_{\mu\nu}$ is symmetric, and $R_{\mu\alpha\nu\beta} g^{\alpha\beta}$ is same as $R_{\mu\alpha\nu\beta} g^{\mu\nu}$, and all other two-index contractions of Reimann tensor are either lead to zero or being anti-symmetric.

Still, there is a problem hidden (below), unless we disclose by taking the covariant derivative as follows:

$$\nabla_\mu R^\mu_\nu \neq 0$$

We use the Bianchi identity,

$$\nabla_{[\rho} R_{\alpha\beta]\mu\nu} = 0$$

Then, upon contraction,

$$g^{\rho\mu} g^{\alpha\beta} \nabla_{[\rho} R_{\alpha\beta]\mu\nu} = 0$$

$$\nabla_{\mu} (R_{\nu}^{\mu} - \frac{\delta_{\nu}^{\mu}}{2} R) = 0$$

Let's define $G_{\nu}^{\mu} = R_{\nu}^{\mu} - \frac{\delta_{\nu}^{\mu}}{2} R$ and note that $\nabla_{\mu} G_{\nu}^{\mu} = 0$ (problem removed) and then, contracting with $g_{\mu\alpha}$ (making things to look beautiful),

$$g_{\mu\alpha} G_{\nu}^{\alpha} = G_{\mu\nu} = g_{\mu\alpha} (R_{\nu}^{\alpha} - \frac{\delta_{\nu}^{\alpha}}{2} R) = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}$$

Therefore, we get the beautiful Einstein's tensor, as

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}$$

For historical reasons¹

Chapter 4

Dynamical Field Equation

4.0.1 Law of Gravitation

The derivation of the Geodesics equation generalises Newton's second law of motion. However, Newton had not only discovered the laws of motion but also provided a mathematical expression of the Law of Gravitation. Our new task is to make similar progress:

- To generalise the RHS of the Law of Gravitation.
- Modify the LHS of Newton's second law of motion to include the idea that gravity is curvature.

Just to get a kind of “Geodesic equation of gravity” (there is no such thing, get it metaphorically) or “Relativistic Newton's law of gravitation” aka Einstein's field equation.

Just like every equation, Newton's law of gravitation also has LHS and RHS.

$$m_{in}\ddot{x} = -G \frac{M_{gr}m_{gr}}{|r_f - r_i|^2}$$

Using the Weak principle of equivalence,

$$\ddot{x} = -G \frac{M_{gr}}{|r_f - r_i|^2} = -\frac{1}{m_{in}} \nabla \phi$$

Recall the fact that conservative $F = -\nabla \phi_g$, where ϕ_g is the gravitational potential. Also, instead of gravitational mass M_{gr} , consider its mass density ρ .

$$\ddot{x} = -G\rho \frac{V}{|r_f - r_i|^2} = -G\rho \frac{4\pi|r_f - r_i|^3}{3|r_f - r_i|^2} = -4\pi G\rho|r_f - r_i|$$

we get,

$$-\nabla^2 \phi = -4\pi G\rho$$

Our aim with LHS is:

$$-\nabla^2 \phi \rightarrow \square g_{00} \rightarrow \eta^{\mu\nu} \partial_\mu \partial_\nu \rightarrow g^{\mu\nu} \nabla_\mu \nabla_\nu \rightarrow R_{\mu\alpha\nu\beta} g^{\alpha\beta} = R_{\mu\nu} \rightarrow G_{\mu\nu} \quad (4.1)$$

Also, our aim with RHS is:

$$-4\pi G\rho \rightarrow -8\pi GT_{00} \rightarrow -8\pi GT_{\mu\nu}$$

Note, $g_{00} = -(1 + 2\phi)$ and $T_{00} = \rho$.

This will end up being

$$G_{\mu\nu} = 8\pi GT_{\mu\nu}$$

(How?): Talking about sequence 4.1. We moved away from $\nabla^2\phi$ and $\square g_{00}$, as they were non-relativistic and not torsionless. Then, we also moved away from $\eta_{\mu\nu}$ since it is only being lorentz invariant but not general coordinate invariant, and then we shifted from $g^{\mu\nu}\nabla_\mu\nabla_\nu$, shows lack of explicite curvature measure, now $R_{\mu\alpha\nu\beta}$, being good choice, is (0,4) type tensor, but not fully symmetric. $R_{\mu\nu}$ is a better choice, but it does not satisfy the Bianchi identity (efforts were made), so modified $G_{\mu\nu}$ is a great choice; it is free from the above issues. Still, it's one of the terms having Λ that has crafted the cosmological constant problem.

In the last chapter, we already got the $G_{\mu\nu}$. Here we argue to get the stress-energy-momentum tensor: $T_{\mu\nu}$. If Lagrangian $\mathcal{L} = \mathcal{L}(\phi, \nabla_\mu\phi)$ then we can find $T_{\mu\nu}$ in its terms.

Let's derive the expression: We start by citing the expression of the conserved four-current j^μ .

$$j^\mu = a^\nu \left[\partial_\nu\phi \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} - \delta_\nu^\mu \mathcal{L} \right]$$

now, calling [...] as T_ν^μ

then

$$j^\mu = a^\nu T_\nu^\mu$$

Since it is being conserved,

$$\partial_\mu a^\nu T_\nu^\mu = 0$$

So,

$$T_\nu^\mu = \partial_\nu\phi \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} - \delta_\nu^\mu \mathcal{L}$$

lowering the index via $T_{\mu\nu} = \eta_{\mu\alpha} T_\nu^\alpha$

then,

$$T_{\mu\nu} = \eta_{\mu\alpha} \partial_\nu\phi \frac{\partial\mathcal{L}}{\partial(\partial_\alpha\phi)} - \eta_{\mu\nu} \mathcal{L}$$

Note that this also satisfies, $\nabla_\mu T^{\mu\nu} = 0$, like $\nabla_\mu G_\nu^\mu = 0$. Also, if you find it difficult to solve the above, one can directly have $T_{\mu\nu}$ as,

$$T_{\mu\nu} = \frac{-2}{\sqrt{-g}} \frac{\delta(\sqrt{-g}\mathcal{L})}{\delta g^{\mu\nu}}$$

Chapter 5

Solve the EEquation

As we constructed the Dynamical field equation (Einstein's field equation), we will try to use that to solve related problems.

Let's consider a static, spherically symmetric matter in space having radius R , so in case of inside the matter, $r < R$, it satisfies $G_{\mu\nu} = 8\pi GT_{\mu\nu}$, otherwise $r \gg R$ (Outside), results that $T_{\mu\nu} = 0 \rightarrow G_{\mu\nu} = 0$.

More elaboration on the outside case,

$$G_{\mu\nu} = 0$$
$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 0$$

Contracting via inverse metric,

$$g^{\mu\nu}R_{\mu\nu} - \frac{1}{2}Rg^{\mu\nu}g_{\mu\nu} = R - \frac{1}{2}R4 = 0$$
$$\implies R = 0$$

Or, $R_{\mu\nu} = 0$ for the outside case. (if $r \gg R$)

In the above calculation, we didn't disclose the form of metric, thus it is valid for all cases, making space-time mostly flat (insignificant curvature) at very distant from matter presence.

5.1 Develop the outside metric and more

Let's ponder over what we mean by static, spherically symmetric. By static, we demand that g_{tt} not have time dependence, and also there should not be g_{it} terms in the line element. Therefore, ∂_t is a Killing vector field, ($\partial_t g_{tt} = 0$ or $\mathcal{L}_t g_{\mu\nu} = 0$). Now you can guess what should be the mathematical description if one demands "Spherically symmetric": $\mathcal{L}_\phi g_{\mu\nu} = 0$, where ϕ is the azimuthal angle, and therefore ∂_ϕ is another Killing vector field (Or, no ϕ dependence in $d\Omega$ term).

This hints that the particular form of metric, such that it holds for $r \gg R$ and results in $R_{\mu\nu} = 0$, should be

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}dr^2 + \tilde{r}^2 d\Omega^2$$

Where $\tilde{r} = re^\gamma(r)$

Our aim is to determine the theses $\alpha(r)$ and $\beta(r)$. We note that only R_{tt} , R_{rr} , $R_{\theta\theta}$, $R_{\phi\phi}$ will be surviving. If one considers this metric and determines the respective Ricci tensor components; Funny enough, then we can explicitly write them (Ricci tensor components and deliberately equate them to zero to get α and β , but the actual reason for equating to zero is $T_{\mu\nu} = 0$) respectively as follows:

- $R_{tt} = e^{2(\alpha-\beta)}[\partial_r^2\alpha + \frac{2}{r}\partial_r\alpha - \partial_r\alpha\partial_r\beta] = 0$
- $R_{rr} = -\partial_r^2\alpha - (\partial_r\alpha)^2 + \partial_r\alpha\partial_r\beta + \frac{2}{r}\partial_r\beta = 0$
- $R_{\theta\theta} = e^{-2\beta}[r(\partial_\beta - \partial_r\alpha) - 1] + 1 = 0$
- $R_{\phi\phi} = \sin^2\theta R_{\theta\theta} = 0$

Which gives [On solving the R_{tt} and R_{rr}],

$$\alpha = -\beta + C$$

And if taking $\tilde{t} = te^C$, then,

$$ds^2 = -e^{-2\beta(r)}d\tilde{t}^2 + e^{2\beta(r)}d\tilde{r}^2 + \tilde{r}^2d\Omega^2$$

On solving the $R_{\theta\theta}$ differential equation, we get $e^{-2\beta} = (1 - \frac{B}{r})$, (it can be shown that this constant B is equal to $2GM/c^2$)

So finally, the outside line element is,

$$ds^2 = \left(1 - \frac{2B}{r}\right)d\tilde{t}^2 + \left(1 - \frac{B}{r}\right)^{-1}d\tilde{r}^2 + \tilde{r}d\Omega^2$$

Therefore, Schwarzschild's solution is the *unique* solution of the vacuum Einstein field equation outside the matter. (Birkhoff's theorem).

We can, for instance, apply this line metric to get the force law of Newton's gravitation and write down the geodesic equation.

$$\ddot{x}^\mu + \Gamma_{\alpha\beta}^\mu \dot{x}^\alpha \dot{x}^\beta = 0$$

Where these non-vanishing components of Christoffel connections are,

$$\begin{aligned} \bullet \Gamma_{tr}^t &= \frac{GM}{r(r-2GM)} & \Gamma_{tt}^r &= \frac{GM}{r^3}(r-2GM) & \Gamma_{rr}^r &= -\frac{GM}{(r-2GM)} \\ \bullet \Gamma_{\theta\theta}^r &= -(r-2GM) & \Gamma_{r\theta}^r &= \Gamma_{r\phi}^\phi = \frac{1}{r} & \Gamma_{\phi\phi}^\theta &= -\sin\theta\cos\theta & \Gamma_{\theta\phi}^\phi &= \cot\theta \end{aligned}$$

5.1.1 Killing vectors and Effective potentials in the Schwarzschild Geometry

So these things are simple and make one perceive the beauty of nature, which is being revealed through mathematical manipulations. We will be doing calculations considering a 2D plane (conserved angular momentum), thus no variation in θ is allowed.

Once you have the Schwarzschild (SW) line element (unit of $G = c = 1$) as,

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2)$$

Just by looking, we see the metric (I hope you know how to construct a metric from a given line element) is independent of time t and azimuthal angle ϕ , so the Killing vector components are $t^\mu = (1, 0, 0, 0)$ and $R^\mu = (0, 0, 0, 1)$; Where we are considering spherical polar coordinate system (t, r, θ, ϕ) , you can therefore convert these Killing vector component to get expressed in any arbitrary coordinate system, then you would not get 1 as a magnitude. For those who still can't see, then note Lie derivative $\mathcal{L}_t g_{\mu\nu} = 0$ and $\mathcal{L}_\phi g_{\mu\nu} = 0$ and the Killing vector fields are ∂_t and ∂_ϕ .

Now, we define magnitude of Energy $E = g_{\mu\nu} t^\mu U^\nu$ and the magnitude of orbital angular momentum $L = g_{\mu\nu} R^\mu U^\nu$, where U_μ is the four proper velocity, with $U^\mu U_\mu = \epsilon$.

Since only $t^0 = 1$ and $R^3 = 1$ are surviving (non-zero), and also noticing metric is diagonal.

$$E = g_{0\nu} t^0 U^\nu = g_{0\nu} U^\nu = g_{00} U^0 = - \left(1 - \frac{2M}{r}\right) \frac{dt}{d\tau}$$

Similarly, [In 2D, $\sin \theta$ is constant]

$$L^\phi = g_{3\nu} R^3 U^\nu = g_{3\nu} U^\nu = g_{33} U^3 = r^2 \sin^2 \theta \frac{d\phi}{d\tau} \approx r^2 \frac{d\phi}{d\tau}$$

Now let's get the effective potential:

We already now

$$g_{\mu\nu} U^\mu U^\nu = \epsilon$$

This $\epsilon = \{-1, 0, 1\}$, corresponds to either timeline, lightlike or spacelike.

Expand the above, [while taking 2D setup, $U^\theta = 0$]

$$\begin{aligned} g_{tt}(U^t)^2 + g_{rr}(U^r)^2 + g_{\phi\phi}(U^\phi)^2 &= \epsilon \\ - \left(1 - \frac{2M}{r}\right) \left(\frac{dt}{d\tau}\right)^2 + \left(1 - \frac{2M}{r}\right)^{-1} \left(\frac{dr}{d\tau}\right)^2 + r^2 \left(\frac{d\phi}{d\tau}\right)^2 &= \epsilon \\ - \left(1 - \frac{2M}{r}\right)^{-1} \left(\frac{dt}{d\tau}\right)^2 + \left(\frac{dr}{d\tau}\right)^2 + \left(1 - \frac{2M}{r}\right) r^2 \left(\frac{d\phi}{d\tau}\right)^2 &= \epsilon \left(1 - \frac{2M}{r}\right) \\ -E^2 + \left(\frac{dr}{d\tau}\right)^2 + \left(1 - \frac{2M}{r}\right) \frac{L^2}{r^2} &= \epsilon \left(1 - \frac{2M}{r}\right) \end{aligned}$$

Upon simplification, we get the general trajectory equation, as

$$\frac{1}{2} \dot{r}^2 + V_{eff} = \tilde{\epsilon}$$

Where, $\tilde{\epsilon} = \frac{E^2}{2}$ and The general effective potential:

$$V_{eff}(r) = \frac{1}{2} \left(1 - \frac{2M}{r}\right) \left(\frac{L^2}{r^2} - \epsilon\right)$$

Which is quite different from Newtonian effective potential (2nd term): $\frac{1}{2}\dot{r}^2 + (-\frac{GM}{r} + \frac{L^2}{m^2 r^2}) = E$. Still, one can show Einstein's agreement at $r \gg \frac{2GM}{c^2}$. For a photon $\epsilon = 0$ and for a massive particle $\epsilon = -1$. We see that there will be a $-\frac{1}{r^3}$ dependent term in the general effective potential, making the orbit of a massive body not completely closed, contrary to classical claims. Additionally, no stable orbit of the photon, but only a photon sphere with radius $R_c = \frac{3GM}{c^2}$. One can do a stability analysis to determine stable and unstable extrema.

What if, introducing a small perturbation?

Given the trajectory equation, we should at least find an unperturbed solution, and if the strength and curiosity remain, we can solve further for the weak perturbation case. The trajectory equation for $\epsilon = 0$ (photon) with gravitational potential $-\frac{2M}{r}$ is,

$$\dot{r}^2 + \left(1 - \frac{2M}{r}\right) \left(\frac{L^2}{r^2}\right) - \frac{2M}{r} = E^2$$

$$\dot{r}^2 + \frac{L^2}{r^2} - \frac{2ML^2}{r^3} - \frac{2M}{r} = E^2$$

We want to see how r changes with ϕ , so using $L = r^2 \frac{d\phi}{d\tau}$,

$$\left(\frac{dr}{d\phi}\right)^2 + \frac{1}{\left(\frac{d\phi}{d\tau}\right)^2} \left(1 + \frac{L^2}{r^2} - \frac{2ML^2}{r^3} - \frac{2M}{r}\right) = \frac{E^2}{\left(\frac{d\phi}{d\tau}\right)^2}$$

Putting value of $\frac{d\phi}{d\tau}$, and taking $x = \frac{L^2}{2mr}$, then,

$$(d_\phi x)^2 + \frac{L^2}{m^2} - 2x + x^2 - \frac{2m^2 x^3}{L^2} = \frac{E^2 L^2}{m^2}$$

acting d_ϕ over the above equation and equating it to zero, gives us,

$$\left(\frac{dx}{d\phi}\right) \left(\frac{d^2 x}{d\phi^2} - 1 + x - \frac{3m^2}{L^2} x^2\right) = 0$$

Which implies either $\frac{dx}{d\phi} = 0$ or,

$$\frac{d^2 x}{d\phi^2} - 1 + x - \frac{3m^2}{L^2} x^2 = 0$$

The solution of the above equation should satisfy the Newtonian trajectory.

$$x = 1 + c \cos \phi + D \sin \phi$$

Therefore, the unperturbed solution should be

$$x = 1 + e \cos(\phi - \phi_0)$$

This is the equation of an elliptical orbit, where e is the eccentricity.

As declared above, we will consider a small perturbation x_1 in $x = x_{Newton} + x_1$, thus, we will be ignoring terms like x_1^2 . Recalling the above differential equation (non-linear) and inserting this perturbed solution, we see,

$$\frac{d^2(x_n + x_1)}{d\phi^2} - 1 + (x_n + x_1) - \frac{3m^2}{L^2}(x_n^2 + 2x_n x_1) = 0$$

$$[\text{Unperturbed non-linear Eq}] + \frac{d^2 x_1}{d\phi^2} - 1 + x_1 - \frac{3m^2}{L^2} \left(1 + 2 \frac{x_1}{x_n} \right) = 0$$

Which has a solution as,

$$x_1 = \frac{3m^2}{L^2} \left[1 + \frac{1}{2}e^2 + e\phi \sin \phi - \frac{1}{6}e^2 \cos 2\phi \right]$$

Note the presence of $e\phi \sin \phi$, causing non-periodic behaviour, leading to a changing location of the extrema. And, the complete solution is,

$$x = x_n + x_1 = 1 + e \cos \phi + \frac{3m^2}{L^2} \left[1 + \frac{1}{2}e^2 + e\phi \sin \phi - \frac{1}{6}e^2 \cos 2\phi \right]$$

On finding $\frac{dx}{d\phi}|_{\phi_0} = 0$ and $\frac{d^2x}{d\phi^2} < 0$, thus maxima at $x \sim \frac{1}{r}$,

$$\implies \sin[(1 - \alpha)\phi_0] = 0$$

where $\frac{3m^2}{L^2} = \alpha$ then, we get,

$$(1 - \alpha)\phi_0 = N\pi$$

Where N is $2n$ for minima and is even, depicts Perihelion, and for maxima, $(2n + 1)$ is odd, represents the case of Aphelion. Here, considering $2n$, with small α ,

$$\implies \phi_0 = \frac{2n\pi}{1 - \alpha} \approx 2n\pi(1 + \alpha)$$

Now the shift is after each cycle of revolution (take $n=1$),

$$\Delta\phi \approx 2\pi\alpha$$

$$\Delta\phi \approx 2\pi \frac{3m^2}{L^2} = 2\pi \frac{3m}{(1 - e^2)a}$$

The above discussed the case around perihelion, where $L^2 = M(1 - e^2)a$, and a is the semi-major axis. For example, the value of $\Delta\phi$ for mercury is $43''$ arc per century.

5.1.2 Removing a singularity through Tortoise

Everyone likes removing apparent difficulties; here, we will be doing a similar exercise.

Again, recall the SW line element,

$$ds^2 = - \left(1 - \frac{2M}{x}\right) d\omega^2 + \left(1 - \frac{2M}{x}\right)^{-1} dx^2 + x^2 d\Omega^2$$

when $x < 2M$,

$$\left(\frac{2M}{|x|} - 1\right) d\omega^2 - \left(\frac{2M}{|x|} - 1\right)^{-1} dx^2 + x^2 d\Omega^2$$

Notice that time acts like space (direction-wise) and vice versa; see the signature has changed from $(-\omega, +x)$ to $(+\omega, -x)$. We conclude, crossing the event horizon $x < 2M$, time and space are interchanged, making the line element as a whole spacelike (causally disconnected); One has to just accept it is a fact. But a more devastating case occurs at $x = 2M$, and this is an apparent problem when one uses the given coordinates. Now, ask can build up a new coordinate to solve the above problem or where the domain of say $r = [2M, \infty)$ can be mapped to $r_* = (-\infty, \infty)$. With potentially r_* , which would not be noticing any shooting up the value, but would remain a casual discontinuity when crossed. There are two kinds of singularities, one occurs at $x = 2M$ and the other at $x = 0$. It has already been stated that the former is just apparent, so it can be removed by choosing an appropriate coordinate system.

Let's try, Take out the problematic factor in the line element as a common factor,

$$ds^2 = - \left(1 - \frac{2M}{r}\right) \left[dt^2 - \left(1 - \frac{2M}{r}\right)^{-2} dr^2 \right] + r^2 d\Omega^2$$

We define a differential of 'new' coordinate

$$dr_* = \left(1 - \frac{2M}{x}\right)^{-1} dr$$

Then, upon indefinite integration and setting the constant of integration $C = -2M \ln 2M$, We end up having

$$r_* = r + 2M \ln \left(\frac{r}{2M} - 1 \right)$$

This is called the **tortoise coordinate system** (t, r_*) . The name is given such, because it makes you feel that covering a finite distance is infinite. Also, no stop to instead have $t_* = t + 2m \ln |r - 2M|$.

5.1.3 Outside and at the event horizon

Now substitute back this r_* into the line element.

$$ds^2 = - \left(1 - \frac{2M}{r}\right) [dt^2 - dr_*^2] + r^2 d\Omega^2$$

Upon factoring $dt^2 - dr_*^2$ and defining each factor as $dU = dt - dr_*$ and $dV = dt + dr_*$. then,

$$ds^2 = - \left(1 - \frac{2M}{r(r_*)} \right) [dU dV] + r^2(r_*) d\Omega^2$$

This now looks compact. Also one if desire to get r_* in terms of U and V , then

$$r_* = \frac{V - U}{2} = r + 2M \ln \left(\frac{r}{2M} - 1 \right)$$

Despite all these manipulations, we have not fully resolved the problem, since $r(r_*)$ is still present in a unique fashion (that keeps the problem alive) in a common factor of ds^2 .

On dividing the above by $2M$,

$$\frac{V - U}{4M} = \frac{r}{2M} + \ln \left(\frac{r}{2M} - 1 \right)$$

Raising by an exponential on both sides,

$$e^{\frac{V-U}{4M}} = e^{\frac{r}{2M}} \left(\frac{r}{2M} - 1 \right)$$

Hold here for a minute!

Meanwhile, let's define

$$V' = e^{V/4M} \quad U' = -e^{-U/4M}$$

Then respectively,

$$dV' = \frac{dV}{4M} e^{V/4M} \quad dU' = \frac{dU}{4M} e^{-U/4M}$$

Then

$$dV' dU' = \frac{dV dU}{16M^2} e^{\frac{V-U}{4M}} = ?$$

Do you know how to replace the above exponential factor $e^{\frac{V-U}{4M}}$? Yes, just lose your 'Hold' and pick that expression and put its value to replace the exponential factor, then,

$$\begin{aligned} \implies dV' dU' &= \frac{dV dU}{16M^2} e^{\frac{r}{2M}} \left(\frac{r}{2M} - 1 \right) \\ \implies dV dU \left(\frac{r}{2M} - 1 \right) &= 16M^2 dV' dU' e^{-r/2M} \end{aligned}$$

Now recall the modified form of ds^2 , so we can further treat its embedded problem. I have written the same below only after taking $2M/r$ out as common,

$$ds^2 = - \frac{2M}{r} \left[\left(\frac{r}{2M} - 1 \right) dU dV \right] + r^2(r_*) d\Omega^2$$

Now, do you identify this term within the square bracket, isn't that equal to $16M^2 dV' dU' e^{-r/2M}$, answer is again yes-yes-yes. Therefore, upon substituting, we finally get, line element for outside and at the event horizon (free from coordinate singularity),

$$ds^2 = - \frac{32M^3}{r(r_*)} dV' dU' e^{-r/2M} + r^2(r_*) d\Omega^2$$

Outside	Inside
$r_* = r + 2M \ln \left(\frac{r}{2M} - 1 \right)$	$R = r - 2M \ln \left \frac{r}{2M} - 1 \right $
$V' = e^{V/4M}$	$\tilde{V} = -e^{-V/4M}$
$U' = -e^{-U/4M}$	$\tilde{U} = e^{U/4M}$

Table 5.1: Kruskal instructions for double null coordinate inside and outside of event horizon.

This is much better than ever before. But only the issue appears as $r \rightarrow 0$ (Curvature singularity). This is also known as the **double null coordinate system** of Schwarzschild geometry. and (U', V') is double null coordinates. Hence, we have successfully treated the coordinate singularity (when $r = 2M$).

5.1.4 Inside event horizon but outside the Black hole

Since using the double null coordinate system, we can cross the event horizon and experience causally disconnected settings; importantly, we have not passed through any singularity point (when we crossed), because our coordinate system is a reinforced beast. Anyway, theoretically, the double null coordinate system allows us to probe inside an event horizon (created by a black hole). Let's start determining the insides of the event horizon.

There is a subtle difference with the toitoise coordinate for the inside case,

$$R = r - 2M \ln \left| \frac{r}{2M} - 1 \right|$$

You should have noted the sign change and consideration of absolute value in comparison with the coordinate for the outside. In fact, this will also modify $U' \rightarrow \tilde{U}$ and $V' \rightarrow \tilde{V}$ form. Let me make think clear to you, basically, there are Kruskal instructions for the inside and outside cases, see the Table 5.1.

With this set of instructions, even the overall form of the line element inside the event horizon would remain the same mathematical form.

$$ds^2 = -\frac{32M^3}{r(R)} e^{-r/4GM} d\tilde{U} d\tilde{V} + r^2 d\Omega^2$$

But why such Kruskal instructions? Because when we cross the event horizon, the $r \rightarrow$ time-like and $t \rightarrow$ spacelike, therefore, in order to accommodate such a change, Kruskal coordinates are of such fashion.

As we did in the outside case, for the inside,

$$dU' = dT - dR \quad dV' = dT + dR$$

Also, $R = \frac{V' - U'}{2}$ then,

$$T^2 - R^2 = \tilde{U} \tilde{V} = -e^{\frac{V-U}{4GM}} = -e^{\frac{r_*}{2GM}}$$

Upon substituting the value of $r - *$, for the inside case, we have

$$T^2 - R^2 = -e^{\frac{r}{2M} - \ln(r-2M)}$$

More compactly, we have written in Table 5.2. You can generate a diagram based on these $T - R$ axes, as Figure 5.1, where a conformal transformation was made to generate the Penrose-Kruskal diagram.

Outside	Inside
$T^2 - R^2 = -e^{\frac{r}{2M}}(r - 2M)$	$R^2 - T^2 = -e^{\frac{r}{2M}}(r - 2M)$

Table 5.2: Kruskal instructions for double null coordinate inside and outside of event horizon.

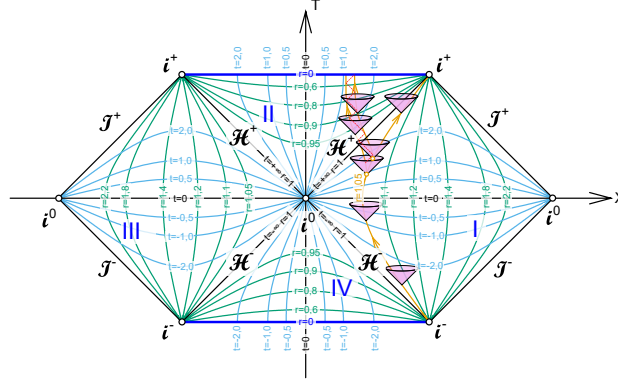


Figure 5.1: The Penrose-Kruskal diagram. (Credit: Wikimedia)

5.2 Interior Solution: Fluid model

Let's keep a happy mind and put more effort into determining the other solution set of Einstein's field equations. We have seen how exterior solution (no matter present, or very far from matter presence) provides a vivid picture of fascinating results about the spacetime fabric, such as the Kruskal diagram of Schwarzschild. Most of the things start from,

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}$$

Note here $T_{\mu\nu} \neq 0$, obviously essential for getting interior solution.

Here we will also be having a line element, but a bit different from the exterior solution, still satisfying the static and spherically symmetric requirements.

$$ds^2 = -e^{2\alpha(r)} dt^2 + e^{2\beta(r)} dr^2 + r^2 d\Omega^2$$

Using this and subsequently determining $g_{\mu\nu}$, $\Gamma_{\mu\nu}^\lambda$, then its derivative, then $R_{\mu\nu}$ then finally $G_{\mu\nu}$, (since, this time $T_{\mu\nu} \neq 0$) we get components of Einstein tensor as follows:

- $G_{tt} = \frac{1}{r^2} e^{2(\alpha-\beta)} (2r\partial_r\beta - 1 + e^{2\beta})$
- $G_{rr} = \frac{1}{r^2} (2r\partial_r\alpha + 1 - e^{2\beta})$
- $G_{\theta\theta} = r^2 e^{-2\beta(r)} [\partial_r^2\alpha + (\partial_r\alpha)^2 - \partial_r\alpha\partial_r\beta + \frac{1}{r}(\partial_r\alpha - \partial_r\beta)]$
- $G_{\phi\phi} = \sin^2\theta G_{\theta\theta}$

Previously, for the exterior solution, we had such a result, but for the Ricci tensor components. Since matter is present, then, let's consider the simplest fluid body (perfect fluid),

$$T_{\mu\nu} = (P + \rho)U_\mu U_\nu + P g_{\mu\nu}$$

Where, P is pressure, ρ is fluid density, and proper velocity $U_\mu = (-e^{\alpha(r)}, 0, 0, 0)$, having $U^\mu U_\mu = -1$. From $T_{\mu\nu}$ of the perfect fluid, we can determine its components using metric $g_{\mu\nu}$ and U_μ ,

We have diagonal elements as,

$$\text{dia}(T_{\mu\nu}) = (\rho e^{2\alpha(r)}, P e^{2\beta(r)}, P r^2, P r^2 \sin^2 \theta)$$

Now on multiplying all diagonal elements by $8\pi G$ should give respective $G_{\mu\nu}$ components, for example $G_{tt} = 8\pi G \rho e^{2\alpha(r)}$.

This allows us to compare the respective Einstein tensor components obtained via the line element with those obtained via the perfect fluid model.

We have,

- $G_{tt} = \frac{1}{r^2} e^{2(\alpha-\beta)} (2r \partial_r \beta - 1 + e^{2\beta}) = 8\pi G \rho e^{2\alpha(r)}$
- $G_{rr} = \frac{1}{r^2} (2r \partial_r \alpha + 1 - e^{2\beta}) = P e^{2\beta(r)}$
- $G_{\theta\theta} = r^2 e^{-2\beta(r)} [\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta + \frac{1}{r} (\partial_r \alpha - \partial_r \beta)] = 8\pi G P r^2$
- $G_{\phi\phi} = \sin^2 \theta G_{\theta\theta} = 8\pi G P r^2 \sin^2 \theta$

What should be done now, except for simplifying the above result as follows:

- $tt : \frac{e^{-2\beta}}{r^2} (2r \partial_r \beta - 1 + e^{2\beta}) = 8\pi G \rho$
- $rr : \frac{e^{-2\beta}}{r^2} (2r \partial_r \alpha + 1 - e^{2\beta}) = 8\pi G P$
- $\theta\theta : e^{-2\beta} [\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta + \frac{1}{r} (\partial_r \alpha - \partial_r \beta)] = 8\pi G P$

Appearance of $e^{-2\beta}$ as a common factor motivates us to make an ansatz, which involves the Misner-Sharp mass $m(r)$, now being an unknown parameter,

$$e^{-2\beta} = \left(1 - \frac{2Gm(r)}{r}\right)$$

Use the ansatz directly and get

$$m(r) = \frac{r}{2G} (1 - e^{-2\beta})$$

And find the rate of $m(r)$ we get,

$$\frac{dm}{dr} = \frac{1}{2G} \left[r \frac{d}{dr} (1 - e^{-2\beta}) + 1 - e^{-2\beta} \right] = \frac{1}{2G} [2r e^{-2\beta} \partial_r \beta + 1 - e^{-2\beta}] = \frac{e^{-2\beta}}{2G} (2r \partial_r \beta - 1 + e^{2\beta})$$

Note the simplified tt term to replace the expression with the bracket.

$$= \frac{e^{-2\beta}}{2G} [2r\partial_r\beta - 1 + e^{2\beta}] = 4\pi G\rho r^2$$

Then, the Misner-Sharp mass in terms of fluid density is,

$$m(r) = 4\pi \int_0^R dr r^2 \rho$$

This is not a real mass, but a fiducial mass(because no metric element is attached), now to have a true mass.

$$M(r) = 4\pi \int_0^R r^2 e^{\beta(r)} \rho(r) dr$$

We can see that always $M(r) > m(r)$. This points out that gravity is giving rise to some kind of “binding energy of the system”.

Now we already know the procedure of solving tt , this will pave the way to solve rr without much complication, like before.

Consider

$$rr : \frac{e^{-2\beta}}{r^2} (2r\partial_r\alpha + 1 - e^{2\beta}) = 8\pi GP$$

Replace $e^{-2\beta}$ by $\left(1 - \frac{2Gm(r)}{r}\right)$,

Then,

$$\left(1 - \frac{2Gm(r)}{r}\right) (2r\partial_r\alpha + 1 - e^{2\beta}) = 8\pi GPr^2$$

$$2r\partial_r\alpha + 1 - e^{2\beta} - \frac{2Gm(r)}{r} (2r\partial_r\alpha + 1 - e^{2\beta}) = 8\pi GPr^2$$

Just aim at $\partial_r\alpha$, then

$$\left(1 - \frac{2Gm(r)}{r}\right) r\partial_r\alpha = \frac{GM + 4\pi GPr^3}{r}$$

Finally,

$$\frac{d\alpha}{dr} = \frac{Gm(r) + 4\pi Gr^3P}{r(r - 2Gm(r))}$$

Now, one may be interested in knowing how the pressure varies with radius. To get the explicit expression, we use the continuity equation, i.e., $\nabla_\mu T^{\mu\nu} = 0$ or solve the $\theta\theta$ expression with a similar approach. Let's take the path of the application of the continuity equation,

$$\nabla_\mu T^\mu_\nu = 0$$

Recall the fact that the perfect fluid is static and isotropic (Spherically symmetric), so $\partial_t\rho = \partial_tP = 0$ and also $\partial_\phi P = \partial_\theta P = 0$. We get,

$$\implies \nabla_\mu T^\mu_r = (\rho + P) \frac{d\alpha}{dr} + \frac{dP}{dr} = 0$$

Upon inserting the result of $\frac{d\alpha}{dr}$, gives,

$$\frac{dP}{dr} = -(\rho + P) \frac{[Gm(r) + 4\pi Gr^3 P]}{r(r - 2Gm(r))}$$

This is the famous Tolman-Oppenheimer-Volkoff (TOV) equation.

Everything looks effortless once solved. In TOV, we note two unknown parameters ρ and the Misner-Sharp mass $m(r)$; however, one can obtain the latter if the former is known. Therefore, consider the case:

$$\rho(r) = \begin{cases} \rho_* & ; r \leq R \\ 0 & ; r > R \end{cases}$$

Then the Misner-Sharp mass,

$$m(r) = \begin{cases} 4\pi \int_0^r r^2 \rho_* dr = \frac{4}{3}\pi \rho_* r^3 & ; r \leq R \\ m(R) & ; r = R \end{cases}$$

It is time to recall the TOV equation and substitute these two values.

Then,

$$\begin{aligned} \frac{dP}{dr} &= -(\rho + P) \frac{[Gm(r) + 4\pi Gr^3 P]}{r(r - 2Gm(r))} \\ &= -(\rho_* + P) \frac{[G\frac{4}{3}\pi \rho_* r^3 + 4\pi Gr^3 P]}{r(r - 2G\frac{4}{3}\pi \rho_* r^3)} = -\frac{4\pi Gr(\rho_* + P)(\rho_* + 3P)}{3\left(1 - \frac{8\pi G\rho_* r^2}{3}\right)} \end{aligned}$$

To obtain the pressure profile, we go for a variable separable method and solve the integral equation.

Then, while remember $M = M(R)$

$$P(r) = \rho_* \left[\frac{R\sqrt{R - 2GM} - \sqrt{R^3 - 2GM r^2}}{\sqrt{R^3 - 2GM r^2} - 3R\sqrt{R - 2GM}} \right]$$

This is the required pressure profile. Now, check what happens at $P(0)$?

We see,

$$P(0) = \rho_* \left[\frac{R\sqrt{R - 2GM} - \sqrt{R^3}}{\sqrt{R^3} - 3R\sqrt{R - 2GM}} \right]$$

This limits that $\sqrt{R^3} - 3R\sqrt{R - 2GM} \neq 0$, then the Buchdahl limit, $R > 2.25GM$, otherwise, the core pressure shoots to infinity.

Now we are at the stage which allows us to get the line element ds^2 by replacing $e^{2\alpha}$ and $e^{2\beta}$ by something more concrete expressions, we already know from the ansatz, that.

$$e^{2\beta} = \left(1 - \frac{2Gm(r)}{r}\right)^{-1}$$

and to get $e^{2\alpha}$, we do

$$1. \text{ Recall } \nabla_\mu T_r^\mu = 0 \implies \frac{d\alpha}{dr} = -\frac{1}{(\rho+P)} \frac{dP}{dr}.$$

2. We already know expression of $P(r)$, which allows us to get $P+\rho = \rho \frac{2\sqrt{1-2GM/R}}{3\sqrt{1-2GM/R}-\sqrt{1-2Gmr^2/R^3}}$
3. Now, solving $d\alpha = -\frac{dp}{\rho+P}$, in limit running from r to R for $\alpha(r)$
4. We get $\alpha(r) - \alpha(R) = \ln \left| \frac{\rho}{\rho+P(r)} \right|$, which converts to $e^{\alpha(r)} = e^{\alpha(R)} \left(\frac{\rho}{\rho+P(r)} \right)$.
5. Applying boundary condition, at $r = R$, $P(R) = 0$ and the interior metric must match the exterior metric of SW. Gives $e^{2\alpha(R)} = 1 - \frac{2GM(R)}{R}$.
6. So, $e^{\alpha(r)} = \sqrt{1 - \frac{2GM(R)}{R}} \left(\frac{\rho}{\rho+P(r)} \right)$,
7. On substituting the value of $\rho/(\rho+P(r))$ from the above point-2, we get

$$e^{2\alpha(r)} = \left[\frac{3}{2} \sqrt{1 - \frac{2GM(R)}{R}} - \frac{1}{2} \sqrt{1 - \frac{2Gm(r)r^2}{R^3}} \right]^2$$

Thus, we have expressed all the required factors in something meaningful. Now inserting those two factors $e^{2\alpha}$ and $e^{2\beta}$ in the line element, we finally get,

$$ds^2 = - \left[\frac{3}{2} \sqrt{1 - \frac{2GM(R)}{R}} - \frac{1}{2} \sqrt{1 - \frac{2Gm(r)r^2}{R^3}} \right]^2 dt^2 + \left(1 - \frac{2Gm(r)}{r} \right)^{-1} dr^2 + r^2 d\Omega^2$$

Note $M(R)$ is not the true mass, but $m(r)$ is, and the temporal part of the line element as dependent on r^2 , thus interior geometry will keep changing. This marks the end of the discussion of the interior solution.

5.3 Gauge and Wave

Sometimes, we do not feel like completing the task that once started with captivating energies, but still, we complete it to uphold the sanctity of the promise. Let's complete the last section of this chapter.

We are already familiar with how introducing a small perturbation in a solution of the elliptical orbit allows us to study the precession of Mercury, where it was done in Newtonian limits. Here, we also bear the mathematical specification by the so-called "weak gravity" or perturbation (fluctuation) on flat spacetime. I mean $|h_{\mu\nu}| \ll 1$, more appropriately,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$$

You can understand that if there is no matter or energy in the universe, then it can be treated approximately flat ($\eta_{\mu\nu}$ only exists).

Before introducing more expressions like $g^{\mu\nu} = \eta^{\mu\nu} + k^{\mu\nu}$ where at leading order, $k^{\rho\sigma} = -\eta^{\rho\mu} h_{\mu\nu} \eta^{\nu\sigma}$ giving as,

$$g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu}$$

Let me reveal the basic idea: *Fluctuations in the spacetime curvature can produce oscillatory tidal effects and fluctuations propagate through space at the speed of light, and that is referred to as*

a *gravitational wave*. And what bears those fluctuations mathematically? The answer is the metric itself. And from where such fluctuations arise, the answer is that those may be either resulting from the “not good” coordinate transformation (diffeomorphism) or the spacetime geometry. So we want to have such coordinate transformations that are gauge-like and do not produce coordinate fluctuations, because they don’t describe gravitational waves; rather, those are misleading pieces of stuff, which one should avoid by use of a filter (gauge).

Now, carrying the above perturbed flat spacetime metric in hand, we see the chain of modifications run within various expressions, in leading order of $h_{\mu\nu}$ or linearised expressions as follows:

Consider, for example, the Christoffel connection

$$\Gamma_{\mu\nu}^{\alpha} = \frac{1}{2}g^{\alpha\lambda}(g_{\nu\lambda,\mu} + g_{\mu\lambda,\nu} - g_{\mu\nu,\lambda})$$

Substituting $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and $g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu}$, then,

$$= \frac{1}{2}(\eta^{\mu\nu} - h^{\mu\nu}) \left(\frac{\partial}{\partial\mu}(\eta_{\lambda\nu} + h_{\lambda\nu}) + \frac{\partial}{\partial\nu}(\eta_{\mu\lambda} + h_{\mu\lambda}) - \frac{\partial}{\partial\lambda}(\eta_{\mu\nu} + h_{\mu\nu}) \right)$$

Upon simplification while ignoring higher order of $h\partial h_{\mu\nu}$, for it being small, we get, Linearised Christoffel symbol as:

$$\Gamma_{\mu\nu}^{(1)\alpha} = \frac{1}{2}\eta^{\alpha\lambda}(h_{\nu\lambda,\mu} + h_{\mu\lambda,\nu} - h_{\mu\nu,\lambda})$$

Now you know the trick, we can have such linearised expressions for the following tensors,

- Linearised Riemann Tensor: $R_{\mu\nu\alpha\beta}^{(1)} = \frac{1}{2}(h_{\mu\beta,\alpha\nu} + h_{\nu\alpha,\beta\mu} - h_{\mu\alpha,\beta\nu} - h_{\nu\beta,\alpha\mu})$
- Linearised Ricci Tensor:

$$R_{\mu\nu}^{(1)} = \frac{1}{2}(\partial_{\sigma}\partial_{\nu}h_{\mu}^{\sigma} + \partial_{\sigma}\partial_{\mu}h_{\nu}^{\sigma} - \partial_{\mu}\partial_{\nu}h - \square h_{\mu\nu})$$

Where, $h = \eta^{\mu\nu}h_{\mu\nu}$ and flat space d’Alembertian $\square = \eta^{\alpha\beta}\partial_{\alpha}\partial_{\beta} = -\partial_t^2 + \partial_x^2$.

- Linearised Ricci scalar $R^{(1)} = \partial_{\mu}\partial_{\nu}h^{\mu\nu} - \square h$
- Linearised Einstein tensor $G_{\mu\nu}^{(1)} = R_{\mu\nu}^{(1)} - \frac{1}{2}\eta_{\mu\nu}R^{(1)}$
- Linearised Einstein’s Field equation as

$$\frac{1}{2} [\partial_{\sigma}\partial_{\nu}h_{\mu}^{\sigma} + \partial_{\sigma}\partial_{\mu}h_{\nu}^{\sigma} - \partial_{\mu}\partial_{\nu}h - \square h_{\mu\nu} - \eta_{\mu\nu}(\partial_{\mu}\partial_{\nu}h^{\mu\nu} - \square h)] = 8\pi GT_{\mu\nu}$$

Suppose we want to test the “loyalty” of these expressions by determining whether they undergo any sharp changes when undergoing the following coordinate transformation (diffeomorphism) or not.

$$x^{\mu} \rightarrow x'^{\mu} = x^{\mu} + \zeta^{\mu}$$

Then the change in metric is depicted by the Lie derivative (note the primed variable is just relabeled),

$$g_{\mu\nu} \rightarrow g_{\mu\nu} + \mathcal{L}_{\zeta^{\mu'}} g_{\mu\nu}$$

Where,

$$\mathcal{L}_{\zeta^\mu} g_{\mu\nu} = \nabla_\mu \zeta_\nu + \nabla_\nu \zeta_\mu$$

This suggests to us that ζ^μ is a Killing vector field.

Now, say

$$\zeta_{\mu\nu} \rightarrow \zeta_{\mu\nu} = g_{\mu\nu} + \nabla_\mu \zeta_\nu + \nabla_\nu \zeta_\mu$$

Comparing this with $\eta_{\mu\nu} + h'_{\mu\nu}$; given, $h'_{\mu\nu} = h_{\mu\nu} + \nabla_\mu \zeta_\nu + \nabla_\nu \zeta_\mu$ and coordinate transformation be,

$$\approx h_{\mu\nu} + \partial_\mu \zeta_\nu + \partial_\nu \zeta_\mu$$

Then,

$$G'_{\mu\nu} \rightarrow G_{\mu\nu} + \text{extra}$$

Now we investigate this extra,

$$\text{extra} = \frac{1}{2} [\partial_\alpha \partial_\nu (\partial_\mu \zeta_\nu + \partial_\nu \zeta_\mu) + \partial_\alpha \partial_\mu (\partial_\nu \zeta_\mu + \partial_\mu \zeta_\nu) - \partial_\mu \partial_\nu h - \square h_{\mu\nu} - \eta_{\mu\nu} (\partial_\alpha \partial_\beta h^{\alpha\beta} - \square h)]$$

As, $h_\mu^\alpha = \eta^{\alpha\nu} h_{\mu\nu}$ also,

$$h_\mu^{\alpha'} = h_\mu^\alpha + \eta^{\alpha\nu} (\partial_\mu \zeta_\nu + \partial_\nu \zeta_\mu) = h_\mu^\alpha + \partial_\mu \zeta^\alpha + \partial^\alpha \zeta_\mu$$

Similarly,

$$h_\nu^{\alpha'} = h_\nu^\alpha + \partial_\nu \zeta^\alpha + \partial^\alpha \zeta_\nu$$

Also,

$$\partial_\alpha \partial_\nu (\partial_\mu \zeta^\alpha + \partial^\alpha \zeta_\mu) = \partial_\alpha \partial_\nu \partial_\mu \zeta^\alpha + \partial_\alpha \partial_\nu \partial^\alpha \zeta_\mu$$

Similarly,

$$\partial_\alpha \partial_\mu (\partial_\nu \zeta^\alpha + \partial^\alpha \zeta_\nu) = \partial_\alpha \partial_\mu \partial_\nu \zeta^\alpha + \partial_\alpha \partial_\mu \partial^\alpha \zeta_\nu$$

So, finally,

$$-\partial_\mu \partial_\nu h - \square h_{\mu\nu} = -(2\partial_\mu \partial_\nu \partial^\alpha \zeta_\alpha) - \partial_\alpha \partial_\nu \partial^\alpha \zeta_\mu - \partial_\alpha \partial_\mu \partial^\alpha \zeta_\nu$$

Thus, extra = 0

Hence,

$$G'_{\mu\nu} = G_{\mu\nu}$$

From above at one thing is clear: the Einstein tensor retains its form $[\delta G_{\mu\nu}^{(1)} = 0]$ under diffeomorphism $[\delta h_{\mu\nu} = \partial_{(\nu} \zeta_{\mu)}]$.

Now we go back to Linearised Einstein's Field equation, where we see that it should have represented 16 differential equations but, fortunately, we have symmetric metric, thus independent elements are just 10, therefore, due symmetric degree of freedom leads us to the ten coupled differential equations of motion for $h_{\mu\nu}$. However, what appears does not always stand true; similarly, with the use of the correct set of Gauge symmetry, we can reduce this counting from ten to just two. The procedure is as follows:

- Since ζ^μ is an arbitrary four vector, being arbitrary gives us control over four quantities of $h_{\mu\nu}$. And this is done via gauge choice, satisfying diffeomorphism $[\delta h_{\mu\nu} = \partial_{(\nu} \zeta_{\mu)}]$; So we are left with 6 coupled differential equations, which need to be solved.

- Let's define to compactify $G_{\mu\nu}^{(1)}$,

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$$

Here, $h = \eta^{\rho\sigma}h_{\rho\sigma}$ and from this we get

$$h_{\mu\nu} = \bar{h}_{\mu\nu} - \eta_{\mu\nu}\bar{h}$$

Where, $\bar{h} = \eta^{\rho\sigma}\bar{h}_{\rho\sigma} = -h$.

- Now using these to rewrite $G_{\mu\nu}^{(1)}$

$$G_{\mu\nu}^{(1)} = \frac{1}{2} [(\partial^\rho \partial_\mu \bar{h}_{\nu\rho} + \partial^\rho \partial_\nu \bar{h}_{\mu\rho}) - \square \bar{h}_{\mu\nu} + \partial_\mu \partial_\nu \bar{h} - \partial_\mu \partial_\nu \bar{h} - \eta_{\mu\nu}(\partial^\rho \partial^\sigma \bar{h}_{\rho\sigma} - \square \bar{h} + \square \bar{h})] = 8\pi G T_{\mu\nu}$$

Keep an eye on the cancellation terms.

- Let invoke de-Donder Gauge (Harmonic Gauge)

$$\partial^\mu \bar{h}_{\mu\nu} = 0$$

Here, we still have six remaining degrees of freedom (meaning we at least need to solve six coupled differential equations), because just the de-Donder gauge is that which satisfies the above-discussed diffeomorphism.

- Henceforth, we see the $G_{\mu\nu}^{(1)}$ reduced to

$$G_{\mu\nu}^{(1)} = -\frac{1}{2}\square \bar{h}_{\mu\nu} \implies \square \bar{h}_{\mu\nu} = -16\pi G T_{\mu\nu}$$

Thus, it is a wave equation (metric wave) with source term; we can also get for $T_{\mu\nu} = 0$, the free space gravitational wave equation as

$$\square \bar{h} = 0$$

Isn't that resemble $\square A^\mu = 0$, in case of the Electromagnetic wave equation, A^μ is the four-potential, and also recall that we used to have the Lorenz gauge $\partial_\mu A^\mu = 0$, satisfying the gauge transformation $A_\mu \rightarrow A_\mu + \partial_\mu f$? Yes, indeed, also you see here $T_{\mu\nu}$ is analogous to four current j^μ of electrodynamics, thus everything seems nicely analogous, a good sign in physics, I regard it as a mark of the beauty within natural law. But, note that the gravitational wave is about $\sim 10^{20}$ suppressed with respect to the electromagnetic wave, still both travel at the universal speed of light, c . Additionally, another difference is that in electromagnetic theory, the nature of transformation (diffeomorphism) is insensitive to spacetime, or it does not have its origin in coordinate transformation, but in gravity; those are spacetime transformations.

Now this gravitation wave equation with source would have a solution expressed in terms of Green's function $\mathcal{G}(\xi, \xi')$,

$$\bar{h}_{\mu\nu}(x) = 2G \int d^4 x' \mathcal{G}(x, x') T_{\mu\nu}(x')$$

We seem to be stuck at six degrees of freedom, but how can that be reduced to two (as claimed)? To do, again recall ζ_μ but here, along with $\square \bar{h} = 0$, we notice that not all gauge transformations

ζ^μ will preserve de-Donder gauge, but only those residual gauge transformations will preserve it, that satisfy the following wave equation,

$$\square \zeta_\mu = 0$$

In easy terms, Note that we had chosen $\zeta^{\mu'}$ such that it satisfies the condition, $\square \zeta^{\mu'} = 0$, even then $\square \bar{h}_{\mu\nu} = 0$ would hold because of $\delta h_{\mu\nu} = \partial_{(\nu} \zeta_{\mu)}$. Hence, this gives us, additional four constraints on the form of harmonic gauge, thus finally we are left with two coupled degrees of freedom, which physically correspond to two modes of polarisation in gravitational waves.

Chapter 6

Cosmology: Friedmann Universe

In the universe full of darkness, light is rare. In cosmology, we aim to study the physics of bodies lost in oblivion and the darkness itself. Here, observation generally talks in the scale of 1000 Mpc. At such a scale, the universe appears isotropic and homogeneous. Note, such behaviour is generally absent when observations are done for less than the scale of a thousand Mpc.

To describe the universe in terms of tools, which are supplied by the general theory of relativity, makes us motivated to build a suitable metric of the universe in the first place. Since the curvature has its mathematical roots in the metric. Additional requirement, we want curvature not to be sensitive to passage of time, and direction, aka “maximally symmetric spacetime”, which respects homogeneity and isotropy. We already got a basic idea of curvature through the list of tensorial objects, which conceptualise the curvature mathematically (Riemann curvature tensor).

Since the Riemann tensor respects the continuity equation, it follows that, for all vectors V^ρ quantified for direction,

$$V^\rho \nabla_\rho R_{\alpha\beta\mu\nu} = 0$$

Which hints us empirically that $R_{\alpha\beta\mu\nu}$ should be simple products of metric of form (while satisfying anti-symmetry nature of the Riemann curvature tensor),

$$\propto (g_{\alpha\mu}g_{\beta\nu} - g_{\beta\mu}g_{\alpha\nu})$$

One can get the proportionality factor as $\frac{R}{n(n-1)}$, where R is the Ricci scalar, and further it also helps to represent the Ricci tensor as,

$$R_{\alpha\mu} = \frac{R}{n}g_{\alpha\mu}$$

One can draw a conclusion, if $R = 0$ then it implies that both $R_{\alpha\beta\mu\nu} = 0$ and $R_{\alpha\mu} = 0$. Therefore, curvature in maximally symmetric spacetime (MSST) is always guided by R . Now, jump back towards getting that metric, which upholds the MSST.

- The required line element should not having any $dt dx^i$ term, otherwise, just on changing $dt \rightarrow -dt$ or $dx \rightarrow -dx$, would change the line element; and we don't want that to occur.

- The g_{00} should only depend on t only and not on x , and we let ‘restricted’ freedom enjoyed by the spatial metric component, $g_{ij}(t, x)$, more correctly, the factor of $dx^i dx^j$, should be sensitive to (t, x)
- Since no direction is special (isotropic), we say that the factor of $dx^i dx^j$, in particular, should be $a^2(t)g_{ij}(x)$. Here, $a^2(t)$ is called the scale factor.
- With all these demands, we construct a line element of the universe in maximally symmetric spacetime,

$$ds^2 = -d\tilde{t}^2 + \tilde{a}^2(t(\tilde{t}))[d\Sigma^2]$$

Where, $d\tilde{t} = \sqrt{g_{00}}dt$ and $d\Sigma^2 = g_{ij}(x)dx^i dx^j$. If one is able to notice, then interestingly, the form of $d\Sigma^2$ does satisfy or resemble the spatial part of the Schwarzschild metric $[e^{2\beta(r)}dr^2 + r^2 d\Omega^2]$.

Now, having the line element, one can indeed get the metric (move on for a moment, from Cartesian to spherical coordinates). Upon comparing with the Schwarzschild metric, $g_{11} = e^{2\beta(r)}$, $g_{22} = r^2$ and $g_{33} = r^2 \sin^2 \theta$. Then using,

$$R_{\alpha\mu} = \frac{R}{n} g_{\alpha\mu}$$

We get,

- $R_{11} = \frac{2}{r} \frac{d\beta}{dr} = 2kg_{11} = 2ke^{2\beta}$
- $R_{22} = e^{-2\beta}(r \frac{d\beta}{dr} - 1) + 1 = 2kg_2 = 2kr^2$
- $R_{33} = \sin^2 \theta R_{22} = 2kg_2 = 2kr^2 \sin^2 \theta$

Where k is a constant parameter related to types of curvature, i.e., if $k = 0 \rightarrow$ flat: Minkowski, $k > 0 \rightarrow$ positive curvature: Spherical, $k < 0 \rightarrow$ negative curvature: Hyperbola. Experimentally, the value of $k \sim 10^{-6}$, is exceptionally weak spatial curvature. Anyway, We can determine the β , on solving R_{11} and R_{22} in terms of k , which gives,

$$\beta = \frac{1}{2} \ln(1 - kr^2)$$

As a result, that metric, which resembled the Schwarzschild metric once, now takes the following form of the Friedmann metric,

$$ds^2 = -d\tilde{t}^2 + a^2\tilde{t} \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right]$$

6.1 Friedmann Equations

Since we have already built the Friedmann metric, we can invoke Einstein’s field equation for the case of the perfect fluid. Let’s ignore k for being negligible, and coming back to the metric represented in Cartesian coordinates, then solving for Ricci tensorial components in terms of scale factor $a(t)$ as,

$$R_{00} = -\frac{3\ddot{a}}{a} \quad R_{11} = R_{22} = R_{33} = \frac{\ddot{a}}{a} + \frac{2\dot{a}^2}{a^2}$$

Using this, we determine Einstein tensor components as,

$$G_{00} = 3 \left(\frac{\dot{a}}{a} \right)^2 = 8\pi G T_{00} \quad G_{ii} = - \left(\frac{\dot{a}}{a} \right)^2 - \frac{\ddot{a}}{a} = 8\pi G T_{ii}$$

Let's say, the Stress-energy tensor $T_{\mu\nu}$ is taking contributions from Baryonic matter, dark matter, photons and so on. It can be modelled for a perfect fluid ($P = \omega\rho$) (homogeneous and isotropic settings) as,

$$T_{\mu\nu} = (P + \rho)U_\mu U_\nu + P g_{\mu\nu}$$

Here, P is pressure, ρ is matter-energy density, U_μ is co-moving observer velocity.

Ultimately, in the co-moving frame, we have the following two equations arising from comparing Einstein tensor components with respective Stress-Energy tensor components. The first is simply as,

$$\left(\frac{\dot{a}}{a} \right)^2 = \frac{8}{3}\pi G T_{00} = \frac{8}{3}\pi G \rho$$

And the second,

$$- \left(\frac{\dot{a}}{a} \right)^2 - \frac{\ddot{a}}{a} = 8\pi G T_{ii} = -\frac{8}{3}\pi G \rho - \frac{\ddot{a}}{a} = 8\pi G ((P + \rho)U_i U_j + P g_{ij})$$

In the co-moving frame,

$$\implies \frac{\ddot{a}}{a} = -\frac{4}{3}\pi G [\rho + 3P]$$

These two are called the first and second Friedmann equations, respectively. What should we do now with these powerful equations? Basically, this implies that the universe is not static. To be static, we should have $\ddot{a}$ and $\dot{a}$ being zeros. But clearly, $\dot{a} \neq 0$, nor $\ddot{a} \neq 0$, and later is negative, provided $\rho > 0$ and $P > 0$. So, should we conclude that the universe is expanding (from the first equation) and, from the second expression, the universe is contracting or decelerating? Before any judgement, we should always talk and see into the one, here seeking cosmological observers (experimentlists) indeed convey to us that the universe is expanding (consistent with the first equation prediction, and measurement done by Hubble in 1928), now this conclusion contradicts our expectation (How?). At this stage, what Einstein would have done is to add Λ to restore the idea of a static universe (to remain consistent with the then prevalent notion of a static universe), before actual measurement done by Hubble, to show the universe is expanding, leading Einstein to accept his "Biggest blunder".

In order to agree with the observation, we postulate a hypothetical fluid, whose contribution would wholly make $(\rho + 3P) < 0$ hugely. Let's say the dispersion relation of a hypothetical fluid is $\rho' = \omega P'$. Then the second Friedmann equation gives,

$$\begin{aligned} \frac{\ddot{a}}{a} &= -\frac{4}{3}\pi G \rho' [1 + 3\omega] > 0 \\ -3\omega > 1 &\implies \omega < -\frac{1}{3} \end{aligned}$$

Unfortunately, we have not yet found any negative ω .

It will be a better practice to write P_i and ρ_i , since we can have contributions from all sorts of fluid sources like baryonic matter, radiation, dark matter, vacuum, etc.

Now, let's make a theoretical correction on the matter side to match experimental results, Given:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \sum_i (\rho_i + 3P_i) \stackrel{demand}{>} 0$$

Which means,

$$\sum_i (\rho_i + 3P_i) < 0 \implies \rho_0 + 3P_0 + \sum_{i \neq 0} (\rho_i + 3P_i) < 0$$

As we know, the density and pressure contributions from all sources except the hypothetical matter are positive, that is $\sum_{i \neq 0} (\rho_i + 3P_i) > 0$

So we demand that $\rho_0 + 3P_0$ is hugely negative,

$$\rho_0(1 + 3\omega_0) < -\sum_{i \neq 0} \rho_i(q + 3\omega_i)$$

On solving for ω_0 , we have,

$$\omega_0 < -\sum_{i \neq 0} \frac{\rho_i}{3\rho_0} (1 + \omega_i) - \frac{1}{3}$$

This is a required condition to agree with the universe's accelerating expansion.

You will be happy to know that one has further known the left side contributions up to when considering both relativistic and non-relativistic ($\frac{3}{7}$) contributions, which is equal to $\frac{1}{2}$, (as per present data- exercise: search, source of data?).

Therefore,

$$\omega_0 < -\frac{1}{2}$$

Which is larger than $\omega_0 < -\frac{1}{3}$, that was determined consider all contributions ($i \neq 0$) become negligible or zero.

6.1.1 On the earliest expansion

Since the universe is expanding, can we know whether it is isothermal expansion, isobaric or adiabatic or other? Yes, we can know. Suppose, earlier universe had only one fluid source, ρ ,

From the 1st Friedmann eq,

$$\rho = \frac{3}{8\pi G} \left(\frac{\dot{a}}{a} \right)^2$$

Differentiate with respect to time,

$$\dot{\rho} = \frac{3}{4\pi G} \left(\frac{\dot{a}}{a^2} \ddot{a} - \frac{\dot{a}^3}{a^3} \right) = \frac{3}{4\pi G} \frac{\dot{a}}{a} \left(\frac{\ddot{a}}{a} - \frac{\dot{a}^2}{a^2} \right)$$

We can again use Friedmann equations, and replace terms like $\frac{\dot{a}}{a}$, $\frac{\ddot{a}}{a}$, then,

$$\dot{\rho} = \frac{3}{4\pi G} \sqrt{\frac{8\pi G \rho}{3}} \left(-\frac{4\pi G \rho}{3} \rho(1+3\omega) - \frac{8\pi G \rho}{3} \right)$$

Upon simplification,

$$\dot{\rho} = -3\frac{\dot{a}}{a}(\rho + P)$$

Now equate it to zero,

$$-3\frac{\dot{a}}{a}(\rho + P) = 0 \implies \dot{\rho} = \frac{d}{dt}(\rho a^3) + \left(\frac{d}{dt}a^3\right)P = 0$$

On multiplying by co-moving volume, $dV = dx dy dz$, and taking $V = a^3 dV$

$$\dot{\rho} = \frac{d}{dt}(\rho a^3 dV) + \left(\frac{d}{dt}a^3 dV\right)P = 0$$

$$\dot{\rho} = \frac{d}{dt}(\rho V) + \left(\frac{d}{dt}V\right)P = 0 \implies dE + PdV = 0$$

This suggests that earlier fluid has been expanding adiabatically. Since,

$$TdS = 0 \implies dS = 0$$

Furthermore, one can also determine the solution of a differential equation, $\dot{\rho} = 0$,
On taking $P = \omega_i \rho$

$$\frac{d}{dt}(\rho a^3) + \left(\frac{d}{dt}a^3\right)\omega_i \rho = 0$$

Which becomes,

$$\dot{\rho} a^3 + 3\dot{a}^2 \rho(1 + \omega_i) = 0 \implies \frac{\dot{\rho}}{\rho} = -3\frac{\dot{a}}{a}(1 + \omega_i)$$

Gives the solution as,

$$\rho = \rho_0 a^{-3(1+\omega_i)}$$

This exponential decay of ρ for different fluids can be seen in the following table 6.1.

Contribution	ω	Variation
Radiation	$\omega = \frac{1}{3}$	$\rho \sim a^{-4}$
Matter	$\omega = 0$	$\rho \sim a^{-3}$
Dark matter	$\omega = -1$	$\rho \sim a^{-0}$

Table 6.1: Shows how different components varied during the adiabatic expansion phase. This also hints that the universe, when born, should have had a very high/rapid expansion or burst, so it would have overcome the inherent negative (deceleration phase).

Let's exactly investigate what happened at $t = t_0$. We take the above solution ρ and feed it into the first Friedmann equation, then,

$$\frac{\dot{a}}{a} = \sqrt{\frac{8\pi G}{3}} \left[\sum_i \rho_{i0} a^{-3(1+\omega_i)} \right]^{1/2}$$

Meanwhile let's define H_0 : Hubble parameter at $t = t_0$ as,

$$H_0 = \sqrt{\frac{8\pi G}{3}} \left[\sum_i \rho_{i0} \right]^{1/2}$$

Then rewriting the working expression in terms of H_0 ,

$$\frac{\dot{a}}{a} = H_0 \frac{\left[\sum_i \rho_{i0} a^{-3(1+\omega_i)} \right]^{1/2}}{\left[\sum_i \rho_{i0} \right]^{1/2}} = H_0 \left[\sum_i \Omega_i a^{-3(1+\omega_i)} \right]^{1/2}$$

Then, setting the equation for determining T,

$$\int_0^1 \frac{da}{a \left[\sum_i \Omega_{i0} a^{-3(1+\omega_i)} \right]^{1/2}} = \int_0^T H_0 dt = H_0 T$$

We split the integral in three parts with limits: $0 - 10^{-4} - (3/7)^{1/3} - 1$ along with placing respective ω_i and Ω_{i0} , which corresponds to the time spent in the Radiation phase (R), Matter phase (N), and dark energy phase (Λ), then,

$$T = \frac{1}{H_0} \left[\int_0^{3.3 \times 10^{-4}} \frac{da}{a [\Omega_{R0} a^{-4}]^{1/2}} + \int_{3.3 \times 10^{-4}}^{(3/7)^{1/3}} \frac{da}{a [\Omega_{N0} a^{-3}]^{1/2}} + \int_{(3/7)^{1/3}}^1 \frac{da}{a [\Omega_{\Lambda0}]^{1/2}} \right]$$

$$\implies T \sim 10^{16}$$

This gives the time taken, in totality, during the three fluid phases.

Chapter 7

Some solved problems

1. **Question:** In a spacetime described by the line element

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

find out the time taken by a photon in traveling dx unit along the x-axis. If the outgoing ray takes more amount of time than in-moving rays for the same dx what can we conclude about the metric components ?

Answer:

So we need to solve this, let say that μ and ν run from $\{0,1,2,3\}$, such that 0 is a temporal component and other three are $\{x, y, z\}$ spatial components.

Since, we are not given with explicit form of metric $g_{\mu\nu}$, so we simply cannot assume that all its non-diagonal elements be zero, and $\text{diag}\{-1, 1, 1, 1\}$. Instead, we stick to basics and as our concern are with x-axis and time, therefore we make a safe assumption that only non-zero metric components are those which involve time and all x-axis dependence components. Then these 8 non-zero metric components are as

$$g_{\mu\nu} = \begin{pmatrix} g_{00} & g_{01} & 0 & 0 \\ g_{10} & g_{11} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

So above given line elements expand as:

$$ds^2 = g_{00}(dx^0)^2 + g_{01}(dx^0)(dx^1) + g_{10}(dx^1)(dx^0) + g_{11}(dx^1)^2$$

Since, we know photon travels at constant c , then dx^0 should be $c dt$ and we may also call dx^1 as dx . Which gives,

$$ds^2 = g_{00}(c dt)^2 + (g_{01} + g_{10}) c dt dx + g_{11}(dx)^2$$

Since, we need to find the time dt , also for photon we know, $ds^2 = 0$.

$$0 = g_{00}(c dt)^2 + (g_{01} + g_{10}) c dt dx + g_{11}(dx)^2$$

This is a quadratic in dt . Solving for dt :

$$dt = \frac{-(g_{01} + g_{10}) \pm \sqrt{(g_{01} + g_{10})^2 - 4g_{00}g_{11}}}{2g_{00}c} dx$$

We take the positive root for outgoing ray. For fixed dx , if $dt_{\text{out}} > dt_{\text{in}}$, then either $g_{01} + g_{10} \neq 0$ or the metric components differ for outgoing vs incoming rays. This suggests the metric is *non-symmetric* ($g_{\mu\nu} \neq g_{\nu\mu}$) or the spacetime is not static (time-reversal asymmetry).

2. **Question** If expanding universe can be explained by a spacetime

$$ds^2 = -c^2 dt^2 + a^2(t) d\mathbf{x}^2$$

Show that if $a(t) = (Ht)^{1/2}$ for a constant H , the distance traveled by photons since the beginning of the universe $t = 0$ is finite. Then find out the condition on n for which the distance traveled will be infinite.

Answer:

We know again that $ds^2 = 0$ for photon, thus

$$c^2 dt^2 = a^2(t) d\mathbf{x}^2$$

$$c dt = a(t) d\mathbf{x}$$

$$c \int_0^t \frac{dt}{a(t)} = \int_0^x d\mathbf{x}$$

Substituting $a(t) = (Ht)^{1/2}$,

$$\begin{aligned} \Rightarrow \int_0^x d\mathbf{x} &= c \int_0^t \frac{dt}{\sqrt{Ht}} \\ \mathbf{x} &= \frac{2c}{\sqrt{H}} \sqrt{t} \end{aligned}$$

So, we see the solution is finite at $t = 0$. In the second part of the question, we are asked to find out the condition on n for which the distance travelled will be infinite.

Let the relation be $a(t) = (Ht)^{n/2}$.

As a result, the integral equation becomes,

$$\int_0^\infty d\mathbf{x} = c \int_0^t \frac{dt}{(Ht)^{n/2}}$$

$$\Rightarrow x = \frac{c}{H^{n/2}} \frac{t^{1-n/2}}{1-n/2}$$

It is clear that if $1 - n/2 \leq 0$, the distance becomes infinite as $t \rightarrow \infty$. So, $n \geq 2$ is the required condition. (For $n = 2$, the integral gives logarithmic divergence.)

3. **Question:** In the Newtonian gravity description, the spacetime line element is

$$ds^2 = -c^2 \left(1 - \frac{2GM}{c^2 r}\right) dt^2 + \left(1 + \frac{2GM}{c^2 r}\right) dr^2 + r^2 d\Omega^2$$

If a photon is moving radially, what is the coordinate velocity it is having in this set-up? Find out the coordinate set-up in which the coordinate velocity is c .

Answer: In the case of a photon, $ds^2 = 0$. For radial motion ($d\Omega = 0$):

$$-c^2 \left(1 - \frac{2GM}{c^2 r}\right) dt^2 + \left(1 + \frac{2GM}{c^2 r}\right) dr^2 = 0$$

Dividing by dt^2 ,

$$v_r = \frac{dr}{dt} = c \sqrt{\frac{1 - \frac{2GM}{c^2 r}}{1 + \frac{2GM}{c^2 r}}}$$

v_r is the required radial velocity.

In the 2nd part of the question, we want $v_r \rightarrow c$. This requires

$$\frac{1 - \frac{2GM}{c^2 r}}{1 + \frac{2GM}{c^2 r}} \rightarrow 1 \quad \Rightarrow \quad \frac{2GM}{c^2 r} \ll 1$$

Thus, $r \gg \frac{2GM}{c^2}$ (far from the gravitating mass). The setup is the weak-field limit, large separation between the photon and the mass.

Also note that radial velocity will become exactly c if we simply take the condition $r^2 d\Omega^2 = 0$ (strictly radial motion in flat space limit).

4. **Question:** In the Newtonian gravity set-up, given above, find out the angle between the radially ingoing ray and out-going ray in the $t - r$ plane. What is expected about this opening angle between the two null rays for the limits $r \rightarrow \infty$ and $r \rightarrow \frac{2GM}{c^2}$?

Answer:

This question itself verifies our conclusion about the setup we imagined in the last question.

Anyway, as per question we can trace a diagram as follows: Both rays are null. In the $t - r$ plane, outgoing ray has $dr/dt = +v_r$, ingoing ray has $dr/dt = -v_r$. The slopes are $\pm 1/v_r$ in the (r, t) plane. The angle between them:

$$\tan \theta = \frac{2v_r/c}{1 - (v_r/c)^2}$$

As $r \rightarrow \infty$: $v_r \rightarrow c$, so $\tan \theta \rightarrow \infty$, i.e., $\theta \rightarrow 90^\circ$.

As $r \rightarrow 2GM/c^2$: $v_r \rightarrow 0$, so $\tan \theta \rightarrow 0$, i.e., $\theta \rightarrow 0^\circ$ (the rays become nearly parallel/tangential).

The expected thing about this is that far from the mass, the light cone opens to 90° (flat spacetime), while near the Schwarzschild radius, the cone collapses, making it hard for light to escape.

5. **Question:** We can not use a photon's proper time for describing its velocity etc. But we can use someone else's proper time to analyse the movement of photons. Using such a clock τ write down the **null** geodesic equation in a static metric. What is the Newtonian acceleration of photons in static spacetime ?

Answer: As photon proper time interval $d\tau = 0$, we cannot use it to parameterize the photon's worldline. Instead we use an affine parameter λ . For a static metric:

$$ds^2 = g_{00}(x)dt^2 + g_{ij}(x)dx^i dx^j$$

The null geodesic equation is:

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} = 0$$

For a radial photon ($x^1 = r$), using an external observer's proper time τ (where $dt/d\tau$ is known), we get the Newtonian acceleration:

$$\frac{d^2 r}{d\tau^2} \approx -\frac{1}{2} g^{11} \partial_r g_{00} \cdot c^2$$

In the weak field limit ($g_{00} \approx -(1 + 2\Phi/c^2)$, $g_{11} \approx 1$), this reduces to $a^r \approx -\partial_r \Phi$, exactly the Newtonian gravitational acceleration. So photons "fall" at the same rate as massive particles in a gravitational field, consistent with the equivalence principle.

6. **Question:** In the case if the spacetime has torsion, i.e. $T_{\mu\nu}^\lambda \equiv \Gamma_{\mu\nu}^\lambda - \Gamma_{\nu\mu}^\lambda \neq 0$ a vector (or a co-vector) fails to comeback to itself when parallelly translated along a closed loop, even in the globally flat spacetime (no curvature).

Answer: This is because torsion modifies the affine connection in a way that introduces a "twisting" effect independent of curvature.

When a vector V^μ is parallel transported around an infinitesimal closed loop spanned by dx^μ and dx^ν , the change in the vector is typically given by the Riemann curvature tensor in the torsion-free case. However, when torsion is present, there is an additional contribution:

$$\delta V^\lambda \sim T_{\mu\nu}^\lambda V^\sigma dx^\mu dx^\nu$$

Even in a globally flat spacetime (where the Riemann tensor vanishes), the presence of torsion ensures that a vector transported around a closed loop does not return to its original form. This is because torsion acts as an intrinsic twist in the affine structure of spacetime, independent of curvature.

7. **Question:** Using the relation $\log \det M = \text{Tr}(\log M)$ for a matrix M , find out the expression for $\delta(-g)^{-n/2}/\delta g^{\mu\nu}$, in terms of $\delta g_{\mu\nu}$ and $\delta g^{\mu\nu}$ with g being the determinant of matrix form of $g_{\mu\nu}$.

Answer:

It is given that $g = \det(g_{\mu\nu})$.

We want to compute

$$\frac{\delta(-g)^{-n/2}}{\delta g^{\mu\nu}}.$$

Proceed as:

$$\frac{\delta(-g)^{-n/2}}{\delta g^{\mu\nu}} = -\frac{n}{2}(-g)^{-n/2-1} \frac{\delta(-g)}{\delta g^{\mu\nu}} = -\frac{n}{2}(-g)^{-n/2-1} \left(-\frac{\delta g}{\delta g^{\mu\nu}} \right)$$

Now we need $\frac{\delta g}{\delta g^{\mu\nu}}$. Using the standard identity (derived from $\ln g = \text{Tr} \ln g_{\mu\nu}$), we have:

$$\frac{\delta g}{\delta g^{\mu\nu}} = g g_{\mu\nu}$$

Thus,

$$\frac{\delta(-g)^{-n/2}}{\delta g^{\mu\nu}} = -\frac{n}{2}(-g)^{-n/2-1} (-g g_{\mu\nu}) = -\frac{n}{2}(-g)^{-n/2} g_{\mu\nu}$$

Therefore,

$$\boxed{\frac{\delta(-g)^{-n/2}}{\delta g^{\mu\nu}} = -\frac{n}{2}(-g)^{-n/2} g_{\mu\nu}}$$

8. **Question:** Show that the Einstein's field equation $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ suggests that all vacuum spacetimes are *Ricci flat*, i.e. $R_{\mu\nu} = 0$. What is the condition if the Einstein's equation gets corrected by a presence of cosmological constant to $G_{\mu\nu} = 8\pi G T_{\mu\nu} + \Lambda g_{\mu\nu}$?

Answer:

In order to show $R_{\mu\nu} = 0$ in case of vacuum spacetimes.

Note in vacuum spacetimes, stress-energy tensor vanishes, $T_{\mu\nu} = 0$.

Then Einstein's equation becomes $G_{\mu\nu} = 0$, i.e.,

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 0$$

$$R_{\mu\nu} = \frac{1}{2}g_{\mu\nu}R$$

Multiplying both sides by $g^{\mu\nu}$,

$$\begin{aligned} g^{\mu\nu}R_{\mu\nu} &= \frac{1}{2}g^{\mu\nu}g_{\mu\nu}R = \frac{1}{2} \cdot 4 \cdot R = 2R \\ \implies R - 2R &= -R = 0 \implies R = 0 \end{aligned}$$

Then $R_{\mu\nu} = 0$. Hence Ricci flat.

In the second part of the question (still $T_{\mu\nu} = 0$), we have

$$\begin{aligned} G_{\mu\nu} &= \Lambda g_{\mu\nu} \\ R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R &= \Lambda g_{\mu\nu} \\ R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}(R + 2\Lambda) &= 0 \end{aligned}$$

Multiplying by $g^{\mu\nu}$:

$$\begin{aligned} g^{\mu\nu}R_{\mu\nu} - \frac{1}{2}g^{\mu\nu}g_{\mu\nu}(R + 2\Lambda) &= R - 2(R + 2\Lambda) = 0 \\ \implies -R - 4\Lambda &= 0 \implies R = -4\Lambda \end{aligned}$$

Then substituting back: $R_{\mu\nu} = \Lambda g_{\mu\nu}$ (not Ricci flat unless $\Lambda = 0$).

9. **Question:** A scalar field has the Lagrangian density as

$$\mathcal{L} = \frac{1}{2\alpha} \sin[\alpha(\partial_\mu\phi)(\partial^\mu\phi)] - V(\phi)$$

Obtain the expression of the stress energy tensor prescribed as

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S[\phi]}{\delta g^{\mu\nu}}$$

where $S[\phi]$ is the action for the scalar field.

Answer: Before doing anything else, let's recall the definition of Action S , as,

$$S[\phi] := \int d^4x \sqrt{-g} \mathcal{L}$$

Our aim is to find the variation of this action with respect to $\delta g^{\mu\nu}$:

$$\frac{\delta S[\phi]}{\delta g^{\mu\nu}} = \frac{\delta}{\delta g^{\mu\nu}} \int d^4x \sqrt{-g} \mathcal{L}$$

Using the product rule,

$$= \int d^4x \frac{\delta\sqrt{-g}}{\delta g^{\mu\nu}} \mathcal{L} + \int d^4x \sqrt{-g} \frac{\delta \mathcal{L}}{\delta g^{\mu\nu}}$$

Now, let's find $\frac{\delta\sqrt{-g}}{\delta g^{\mu\nu}}$:

$$\frac{\delta\sqrt{-g}}{\delta g^{\mu\nu}} = -\frac{1}{2}\sqrt{-g} g_{\mu\nu}$$

(This is a standard result; it follows from $\delta g = g g_{\mu\nu} \delta g^{\mu\nu}$.)

Also, find $\frac{\delta \mathcal{L}}{\delta g^{\mu\nu}}$:

$$\frac{\delta \mathcal{L}}{\delta g^{\mu\nu}} = \frac{1}{2} \cos[\alpha(\partial_\mu \phi)(\partial^\mu \phi)] (\partial_\mu \phi)(\partial_\nu \phi)$$

Now substituting these found values, we get:

$$\begin{aligned} \frac{\delta S[\phi]}{\delta g^{\mu\nu}} &= \int d^4x \left[-\frac{1}{2}\sqrt{-g} g_{\mu\nu} \mathcal{L} + \sqrt{-g} \cdot \frac{1}{2} \cos[\alpha(\partial_\mu \phi)(\partial^\mu \phi)] (\partial_\mu \phi)(\partial_\nu \phi) \right] \\ &= \int d^4x \sqrt{-g} \left[-\frac{1}{2} g_{\mu\nu} \mathcal{L} + \frac{1}{2} \cos[\alpha(\partial_\mu \phi)(\partial^\mu \phi)] (\partial_\mu \phi)(\partial_\nu \phi) \right] \end{aligned}$$

Now, $T_{\mu\nu}$ is given by:

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S[\phi]}{\delta g^{\mu\nu}} = -2 \left[-\frac{1}{2} g_{\mu\nu} \mathcal{L} + \frac{1}{2} \cos[\alpha(\partial_\mu \phi)(\partial^\mu \phi)] (\partial_\mu \phi)(\partial_\nu \phi) \right]$$

$$T_{\mu\nu} = g_{\mu\nu} \mathcal{L} - \cos[\alpha(\partial_\mu \phi)(\partial^\mu \phi)] (\partial_\mu \phi)(\partial_\nu \phi)$$

Explicitly substituting $\mathcal{L}$:

$$T_{\mu\nu} = \frac{g_{\mu\nu}}{2\alpha} \sin[\alpha(\partial_\mu \phi)(\partial^\mu \phi)] - g_{\mu\nu} V(\phi) - \cos[\alpha(\partial_\mu \phi)(\partial^\mu \phi)] (\partial_\mu \phi)(\partial_\nu \phi)$$

10. **Question:** A simple spacetime has a certain quantity called geodesic deviation expansion parameter θ defined as

$$\frac{d\theta}{d\lambda} = -\alpha\theta^2 - R_{\mu\nu}u^\mu u^\nu$$

where $R_{\mu\nu}$ is the Ricci tensor, u^μ is tangent along the geodesic which is parameterized by λ and α is a constant. Find out for the geodesic expansion rate to become positive (i.e. for gravity to become repulsive), what is the necessary condition for the matter stress energy tensor $T_{\mu\nu}$, in Einstein's theory of relativity ?

Answer: We aim to show that $\frac{d\theta}{d\lambda} > 0$ for gravity to become repulsive.

That is,

$$-\alpha\theta^2 - R_{\mu\nu}u^\mu u^\nu > 0$$

$$\begin{aligned}
-R_{\mu\nu}u^\mu u^\nu &> \alpha\theta^2 \geq 0 \\
\implies R_{\mu\nu}u^\mu u^\nu &< 0
\end{aligned}$$

Now, we recall Einstein's field equation:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi GT_{\mu\nu}$$

Multiply both sides by $u^\mu u^\nu$:

$$\begin{aligned}
R_{\mu\nu}u^\mu u^\nu - \frac{1}{2}g_{\mu\nu}u^\mu u^\nu R &= 8\pi GT_{\mu\nu}u^\mu u^\nu \\
\implies R_{\mu\nu}u^\mu u^\nu &= 8\pi G \left(T_{\mu\nu}u^\mu u^\nu - \frac{1}{2}Tg_{\mu\nu}u^\mu u^\nu \right)
\end{aligned}$$

So the condition $R_{\mu\nu}u^\mu u^\nu < 0$ becomes:

$$8\pi G \left(T_{\mu\nu}u^\mu u^\nu - \frac{1}{2}T \right) < 0$$

(using $g_{\mu\nu}u^\mu u^\nu = 1$ for timelike geodesics; for null, it's 0 but similar reasoning applies).

Thus, for repulsive gravity, we need:

$$T_{\mu\nu}u^\mu u^\nu < \frac{1}{2}T$$

This violates the *strong energy condition* ($T_{\mu\nu}u^\mu u^\nu \geq \frac{1}{2}T$), indicating that exotic matter (such as with negative pressure or a cosmological constant) is required.

11. **Question:** In the Schwarzschild spacetime given by the line element

$$ds^2 = -c^2 \left(1 - \frac{2GM}{c^2 r} \right) dt^2 + \left(1 - \frac{2GM}{c^2 r} \right)^{-1} dr^2 + r^2 d\Omega^2$$

Express the Killing vector $R = (0, 0, 0, 1)$ in cartesian coordinates (t, x, y, z) . Prove that $(0, z, 0, -x)$ and $(0, 0, -z, y)$ are also Killing vectors.

Answer:

We know that if spacetime is symmetric along the integral curve of a vector field K , and $\mathcal{L}_K g_{\mu\nu} = 0$, then K is a Killing vector field (KVF). If $g_{\mu\nu}$ are independent of one coordinate x^0 , then $\partial/\partial x^0$ is a KVF.

From the Schwarzschild line element, the metric is independent of t and ϕ . The question mentions $R = (0, 0, 0, 1)$ as a KVF, which is ∂_ϕ .

Now, expressing ∂_ϕ in Cartesian coordinates requires the transformation from spherical to Cartesian:

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$

The transformation rule for a vector (1,0) gives:

$$\frac{\partial}{\partial \phi} = \frac{\partial x}{\partial \phi} \frac{\partial}{\partial x} + \frac{\partial y}{\partial \phi} \frac{\partial}{\partial y} + \frac{\partial z}{\partial \phi} \frac{\partial}{\partial z}$$

$$\frac{\partial x}{\partial \phi} = -r \sin \theta \sin \phi = -y, \quad \frac{\partial y}{\partial \phi} = r \sin \theta \cos \phi = x, \quad \frac{\partial z}{\partial \phi} = 0$$

Thus,

$$\frac{\partial}{\partial \phi} = -y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}$$

Therefore, the transformed components of the KVF in Cartesian system are $R = (0, -y, x, 0)$ (the 0 in the time component).

Now we also need to prove that $(0, z, 0, -x)$ and $(0, 0, -z, y)$ are Killing vectors. These correspond to $z\partial_x - x\partial_z$ and $-z\partial_y + y\partial_z$, i.e., rotations about the y -axis and x -axis respectively. Since the Schwarzschild metric is spherically symmetric (independent of rotations), all three rotation generators are Killing vectors. One can verify $\mathcal{L}_K g_{\mu\nu} = 0$ for each explicitly.

For $(0, z, 0, -x)$: $K^1 = z$, $K^3 = -x$. Then:

$$\mathcal{L}_K g_{\mu\nu} = K^\alpha \partial_\alpha g_{\mu\nu} + g_{\alpha\nu} \partial_\mu K^\alpha + g_{\mu\alpha} \partial_\nu K^\alpha = 0$$

because the metric depends only on r , and the derivatives combine to cancel. Similarly for $(0, 0, -z, y)$. Thus, all three are KVF's corresponding to the three independent rotations, confirming the spherical symmetry of Schwarzschild spacetime.

12. **Question:** In Schwarzschild spacetime, if a null ray starts from location (r, θ, ϕ) and moves radially, how much of frequency shift it would undergo when it lands up at (r_0, θ, ϕ) ? Find out the weak field limit of that. How does it scale with the *Newtonian* spatial distance of the points?

Answer:

We know Schwarzschild metric as,

$$ds^2 = - \left(1 - \frac{2GM}{c^2 r} \right) c^2 dt^2 + \left(1 - \frac{2GM}{c^2 r} \right)^{-1} dr^2 + r^2 d\Omega^2$$

For a static observer at radius r , proper time $d\tau = \sqrt{1 - \frac{2GM}{c^2 r}} dt$. For a photon emitted at r with frequency ν (measured locally) and received at r_0 with frequency ν_0 :

Since dt is the same coordinate time interval for emission and reception (for a static spacetime),

$$\nu d\tau = \text{constant} \implies \nu \sqrt{1 - \frac{2GM}{c^2 r}} = \nu_0 \sqrt{1 - \frac{2GM}{c^2 r_0}}$$

Thus,

$$\frac{\nu_0}{\nu} = \sqrt{\frac{1 - \frac{2GM}{c^2 r}}{1 - \frac{2GM}{c^2 r_0}}}$$

The frequency shift is:

$$\frac{\nu_0 - \nu}{\nu} = \sqrt{\frac{1 - \frac{2GM}{c^2 r}}{1 - \frac{2GM}{c^2 r_0}}} - 1$$

In the weak field limit ($GM/(c^2 r) \ll 1$, $GM/(c^2 r_0) \ll 1$), expand:

$$\sqrt{1 - \frac{2GM}{c^2 r}} \approx 1 - \frac{GM}{c^2 r}, \quad \sqrt{1 - \frac{2GM}{c^2 r_0}} \approx 1 - \frac{GM}{c^2 r_0}$$

Then,

$$\begin{aligned} \frac{\nu_0}{\nu} &\approx \frac{1 - \frac{GM}{c^2 r}}{1 - \frac{GM}{c^2 r_0}} \approx \left(1 - \frac{GM}{c^2 r}\right) \left(1 + \frac{GM}{c^2 r_0}\right) \approx 1 + GM \left(\frac{1}{c^2 r_0} - \frac{1}{c^2 r}\right) \\ \frac{\nu_0 - \nu}{\nu} &\approx \frac{GM}{c^2} \left(\frac{1}{r_0} - \frac{1}{r}\right) = \frac{GM}{c^2} \frac{r - r_0}{rr_0} \end{aligned}$$

For small separation $L = r_0 - r$ (with $r_0 > r$), $r_0 \approx r$, then:

$$\frac{\Delta\nu}{\nu} \approx -\frac{GM}{c^2} \frac{L}{r^2}$$

Thus, the frequency shift scales linearly with the Newtonian spatial distance L and inversely with the square of the radial coordinate r .

13. **Question:** In the Kruskal double null coordinate system, identify the points $r = 0$ and $r = \frac{2GM}{c^2}$. Inside the horizon of a black hole, show that the constant spatial coordinate trajectories will also meet the singularity at $r = 0$ in finite proper time.

Answer: We know that the Kruskal coordinates (U', V') are defined by:

$$U' = -e^{-u/4M}, \quad V' = e^{v/4M}$$

where $u = t - r_*$, $v = t + r_*$, and $r_* = r + 2M \ln\left(\frac{r}{2M} - 1\right)$ (in units $G = c = 1$, $M = GM/c^2$).

A key relation is:

$$U'V' = -\left(\frac{r}{2M} - 1\right) e^{r/2M}$$

From this:

- At the event horizon $r = 2M$: $\frac{r}{2M} - 1 = 0$, so $U'V' = 0$.
- At the singularity $r = 0$: $\frac{r}{2M} - 1 = -1$, $e^{r/2M} = 1$, so $U'V' = -1$.

Now for the second part: inside the horizon ($r < 2M$), r becomes timelike and t becomes spacelike. Consider an observer staying at constant spatial coordinates (constant t , θ , ϕ). The line element inside becomes:

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2$$

Since $1 - \frac{2M}{r} < 0$, we can write $\left(\frac{2M}{r} - 1\right) > 0$. Then for a radial trajectory ($d\theta = d\phi = 0$):

$$-d\tau^2 = -\left(\frac{2M}{r} - 1\right)^{-1} dr^2 \implies d\tau = \frac{dr}{\sqrt{\frac{2M}{r} - 1}}$$

The proper time to fall from the horizon ($r = 2M$) to the singularity ($r = 0$) is:

$$\tau = \int_{2M}^0 \frac{dr}{\sqrt{\frac{2M}{r} - 1}} = \frac{\pi M}{2} \quad (\text{finite})$$

Thus, even if you're sitting still (not moving in any spatial direction), you will hit the singularity at $r = 0$ in a finite amount of your own proper time.

14. **Question:** A neutrino is expected to not interact with matter and only move along geodesics provided by the geometry. Inside a star of uniform density, governed by the TOV equation, write down the trajectory of neutrino with energy and angular momentum parameters E and L respectively, and identify the effective potential it experiences.

Answer:

A neutrino moves at the speed of light, so it follows null geodesics ($ds^2 = 0$). The interior solution line element (from the TOV solution) is:

$$ds^2 = -e^{2\alpha(r)} dt^2 + e^{2\beta(r)} dr^2 + r^2 d\Omega^2$$

with $e^{2\beta(r)} = \left(1 - \frac{2Gm(r)}{c^2 r}\right)^{-1}$ and $e^{2\alpha(r)}$ given by the matching to the exterior Schwarzschild solution.

For a null geodesic in the equatorial plane ($d\theta = 0$), using an affine parameter λ :

$$0 = -e^{2\alpha(r)} \left(\frac{dt}{d\lambda}\right)^2 + e^{2\beta(r)} \left(\frac{dr}{d\lambda}\right)^2 + r^2 \left(\frac{d\phi}{d\lambda}\right)^2$$

From the Killing vectors ∂_t and ∂_ϕ , we have conserved quantities:

$$E = e^{2\alpha(r)} \frac{dt}{d\lambda}, \quad L = r^2 \frac{d\phi}{d\lambda}$$

Substitute into the null condition:

$$\begin{aligned} -e^{-2\alpha(r)} E^2 + e^{2\beta(r)} \left(\frac{dr}{d\lambda}\right)^2 + \frac{L^2}{r^2} &= 0 \\ \left(\frac{dr}{d\lambda}\right)^2 &= e^{-2\beta(r)} \left(e^{-2\alpha(r)} E^2 - \frac{L^2}{r^2}\right) \end{aligned}$$

But $e^{-2\beta(r)} = 1 - \frac{2Gm(r)}{c^2 r}$. Therefore,

$$\left(\frac{dr}{d\lambda}\right)^2 = \left(1 - \frac{2Gm(r)}{c^2 r}\right) \left(e^{-2\alpha(r)} E^2 - \frac{L^2}{r^2}\right)$$

We can write this as:

$$\left(\frac{dr}{d\lambda}\right)^2 = \frac{E^2}{e^{2\alpha(r)}} \left(1 - \frac{2Gm(r)}{c^2 r}\right) - \left(1 - \frac{2Gm(r)}{c^2 r}\right) \frac{L^2}{r^2}$$

The effective potential for the radial motion is thus:

$$V_{\text{eff}}(r) = \left(1 - \frac{2Gm(r)}{c^2 r}\right) \frac{L^2}{r^2} - \frac{E^2}{e^{2\alpha(r)}} \left(1 - \frac{2Gm(r)}{c^2 r}\right)$$

(Note: sign conventions vary; often one defines $\dot{r}^2 = E^2 - V_{\text{eff}}$ with a different sign for V_{eff} .)

15. **Question:** For a two fluid model of a star, where constant density ρ fluid occupies interior core $r < R_c$ and the remaining portion $R_c < r < R_*$ is occupied by a second fluid of a different constant density ρ^* , develop the TOV equations. What is the solution for a shell (whose core is empty)?

Answer:

This problem says that

$$\rho(r) = \begin{cases} \rho & r < R_c \\ \rho^* & R_c < r < R_* \end{cases}$$

Let's recall the TOV equation:

$$\frac{dP}{dr} = -(\rho + P) \frac{[Gm(r) + 4\pi Gr^3 P]}{r(r - 2Gm(r))}$$

and

$$m(r) = 4\pi \int_0^r r'^2 \rho(r') dr'$$

Now for the **inner region** ($0 \leq r < R_c$):

$$m(r) = 4\pi \int_0^r r'^2 \rho dr' = \frac{4}{3} \pi \rho r^3$$

Then the TOV equation becomes:

$$\begin{aligned} \left. \frac{dP}{dr} \right|_{\text{in}} &= -(\rho + P) \frac{[G \frac{4}{3} \pi \rho r^3 + 4\pi Gr^3 P]}{r(r - 2G \frac{4}{3} \pi \rho r^3)} \\ &= -4\pi Gr \frac{(\rho + P)(\rho + 3P)}{3(1 - \frac{8}{3} \pi G \rho r^2)} \end{aligned}$$

This can be solved (as in standard constant-density star) to get $P(r)$.

For the **outer region** ($R_c < r < R_*$):

$$m(r) = 4\pi \int_0^{R_c} r'^2 \rho dr' + 4\pi \int_{R_c}^r r'^2 \rho^* dr' = M_c + \frac{4}{3} \pi \rho^* (r^3 - R_c^3)$$

where $M_c = \frac{4}{3}\pi\rho R_c^3$ is the core mass.

Substitute into TOV:

$$\left. \frac{dP}{dr} \right|_{\text{out}} = -(\rho^* + P) \frac{[G(M_c + \frac{4}{3}\pi\rho^*(r^3 - R_c^3)) + 4\pi Gr^3 P]}{r(r - 2G(M_c + \frac{4}{3}\pi\rho^*(r^3 - R_c^3)))}$$

This is a first-order ODE for $P(r)$ that must be solved numerically in general.

Now for a **shell with empty core** ($\rho = 0$, $R_c > 0$, and we consider $r > R_c$): In this case $M_c = 0$, so for $R_c < r < R_*$:

$$m(r) = \frac{4}{3}\pi\rho^*(r^3 - R_c^3)$$

The TOV equation becomes:

$$\frac{dP}{dr} = -(\rho^* + P) \frac{[\frac{4}{3}\pi G\rho^*(r^3 - R_c^3) + 4\pi Gr^3 P]}{r(r - \frac{8}{3}\pi G\rho^*(r^3 - R_c^3))}$$

For a thin shell or in the limit where $R_c \rightarrow 0$, this reduces to the standard constant-density TOV equation. The solution for $P(r)$ can be obtained analytically in terms of the Buchdahl variables, leading to a known closed form (like the one for constant density but with a shifted mass function).